

# AMG Wind Farm

## Power Forecasting System

### Technical Project Report

| Project         | Multi-Horizon Wind Power Forecasting              |
|-----------------|---------------------------------------------------|
| Farm Capacity   | 26.4 MW (12 x 2,200 kW Turbines)                  |
| Raw Data Period | 2021-01-01 to 2026-12-07                          |
| Models          | XGBoost + LightGBM (130 ML models, 520 artifacts) |
| Horizons        | 10 min, 30 min, 1 h, 6 h, 24 h                    |
| API Framework   | FastAPI + Uvicorn (25 endpoints)                  |
|                 |                                                   |
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# 1. Executive Summary

This report presents the design, implementation, and evaluation of a machine learning-based power forecasting system for the AMG Wind Farm. The farm consists of 12 turbines with a total installed capacity of 26.4 MW, located in Vietnam. The system provides multi-horizon forecasts at 10-minute, 30-minute, 1-hour, 6-hour, and 24-hour intervals using XGBoost and LightGBM models.

**Key Results:** The best-performing model achieves  $R^2 = 0.9703$  (farm\_total\_power\_target\_10min (ridge, 10min)). The system includes **130 ML models** (130 = 13 targets x 2 algorithms x 5 horizons, counted from the actual \*\_model.joblib files), **10 baseline evaluations** (walk-forward persistence + ridge), and **520 model artifacts** (model + scaler + features + metadata per trained key). A FastAPI-based REST API serves 25 endpoints including real-time prediction, evaluation metrics, and alert generation. The system writes every output file according to the project output schema (Section 15: Dinh dang file dau ra) with defined column names, forecast\_quality labels, and confidence-interval fields (calibration assessed in Section 5.10, not assumed at a nominal level).

## Champion model per horizon — turbine-avg (min mean RMSE).

The champion is the model with the lowest mean RMSE within each horizon × level cell, computed directly from *evaluation\_metrics.csv*; all four models compete (LightGBM, XGBoost, Ridge, persistence). The best ML model (LightGBM or XGBoost) per cell is listed for comparison.

| Horizon | Level   | Champion | RMSE (kW) | R2     | Best ML  | Best ML RMSE (kW) |
|---------|---------|----------|-----------|--------|----------|-------------------|
| 10min   | Turbine | Ridge    | 146.2     | 0.9419 | LightGBM | 154.2             |
| 30min   | Turbine | Ridge    | 239.4     | 0.8442 | LightGBM | 246.3             |
| 1hour   | Turbine | Ridge    | 306.1     | 0.7453 | LightGBM | 316.1             |
| 6hour   | Turbine | LightGBM | 453.7     | 0.4374 | LightGBM | 453.7             |
| 24hour  | Turbine | Ridge    | 490.5     | 0.3443 | LightGBM | 511.2             |

## Champion model per horizon — farm-total (min mean RMSE).

The champion is the model with the lowest mean RMSE within each horizon × level cell, computed directly from *evaluation\_metrics.csv*; all four models compete (LightGBM, XGBoost, Ridge, persistence). The best ML model (LightGBM or XGBoost) per cell is listed for comparison.

| Horizon | Level | Champion | RMSE (kW) | R2     | Best ML  | Best ML RMSE (kW) |
|---------|-------|----------|-----------|--------|----------|-------------------|
| 10min   | Farm  | Ridge    | 1185.9    | 0.9703 | XGBoost  | 1209.7            |
| 30min   | Farm  | Ridge    | 2303.2    | 0.8880 | LightGBM | 2347.7            |
| 1hour   | Farm  | Ridge    | 3117.4    | 0.7947 | LightGBM | 3198.7            |
| 6hour   | Farm  | LightGBM | 4945.6    | 0.4797 | LightGBM | 4945.6            |
| 24hour  | Farm  | Ridge    | 5516.8    | 0.3525 | LightGBM | 5789.1            |

**v2.1.0 revision note:** This revision addresses the reviewer comments on report v2.0.0 (see Appendix A). Key changes: (1) data coverage is now reported on the observed raw timestamps (292,447 unique rows, 19,601 missing timestamps, 19,601 synthetic rows from 10-minute reindexing); (2) baseline training is

leakage-free and skill scores are computed on identical sample sets with explicit baselines; (3) a leakage audit, provenance inventory, sample trace, and farm-level bias analysis were added; (4) the API authenticates with an environment-variable API key (fail-closed) and CORS is restricted; (5) figure numbering is fixed.

| Metric                  | Value                                                  |
|-------------------------|--------------------------------------------------------|
| Total Turbines          | 12 (TB01 - TB12)                                       |
| Total Capacity          | 26.4 MW                                                |
| Raw Timestamps (unique) | 292,447                                                |
| Raw Missing Timestamps  | 19,601                                                 |
| Models Trained          | 130 ML models (13 targets x 2 algorithms x 5 horizons) |
| Baseline Evaluations    | 10 (walk-forward persistence + ridge)                  |
| Model Artifacts         | 520 (model + scaler + features + metadata per key)     |
| Best R2                 | 0.9703 (farm_total_power_target_10min (ridge, 10min))  |
| Avg Availability        | 82.17%                                                 |
| Output Files            | 26 CSV files (Section 15 output schema)                |
| API Endpoints           | 25 (FastAPI + Uvicorn)                                 |
| API Tests               | 25 passing tests                                       |

## 2. Project Overview

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### 2.1 Wind Farm Description

The AMG Wind Farm is a 26.4 MW wind power facility located in Vietnam, comprising 12 Vestas-class wind turbines (TB01 through TB12), each rated at 2,200 kW. Key turbine specifications:

| Parameter          | Value  | Unit    |
|--------------------|--------|---------|
| Rated Power        | 2,200  | kW      |
| Cut-in Speed       | 3.0    | m/s     |
| Rated Speed        | 12.0   | m/s     |
| Cut-out Speed      | 25.0   | m/s     |
| Sampling Interval  | 10     | minutes |
| Number of Turbines | 12     | -       |
| Total Capacity     | 26,400 | kW      |

### 2.2 Project Objectives

- Develop a multi-horizon power forecasting system for individual turbines and the entire farm
- Implement and compare XGBoost and LightGBM gradient boosting models
- Build a 15-step automated pipeline from raw SCADA data to forecast output
- Create a REST API with interactive dashboard for real-time forecasting
- Detect ramp events, anomalies, and turbine failure risks
- Produce output files following the project output schema (Section 15: Dinh dang file dau ra)
- Achieve  $R^2 > 0.90$  at the 10-minute horizon for individual turbines

## 3. Data Description

### 3.1 SCADA Data Overview

The dataset consists of 11 semi-annual Excel files with overlapping coverage. Raw SCADA records span **2021-01-01 to 2026-12-07** at a nominal 10-minute sampling interval. After unioning all files and de-duplicating timestamps, the observed coverage is **292,447 unique timestamps** out of 312,048 expected steps over that span (coverage 93.72%; 19,601 missing timestamps; 0 duplicate rows removed). Each file contains 49 columns across 12 turbines.

**Important data caveats (reported for transparency):** (1) the raw source files extend beyond the report date (raw union end 2026-12-07), so the official evaluation window is explicitly cut at **evaluation\_cutoff = min(report\_date, raw\_union\_end) = 2026-08-01** (report date 2026-08-01). All test rows at/after the cutoff are flagged **is\_simulated=1** in the processed data and are **excluded from the official evaluation** (18,576 rows set aside); every metric in this report is computed only on the official window (test window ends 2026-07-31). This is a supplier-side file mislabel: the last source file is named [01.2026-07.2026.xlsx](#) but its rows extend to 2026-12-07; the raw files are ingested as-is and the cutoff truncation is the pipeline's defense (flagged to the export owner, see README). (2) to obtain a regular 10-minute grid the pipeline re-indexes the timestamp axis, which introduces **19,601 synthetic rows** out of 312,048 processed timestamps (synthetic ratio 6.28%) — these rows are forward-filled with observed values and flagged **is\_synthetic=1**; any row whose value was forward-filled is flagged **is\_imputed=1**. Rows whose target is synthetic or imputed are **excluded from the official evaluation metrics** (Section 5.1) and from the leakage audit (Section 4.4).

| Coverage Item                              |                                       | Value                                                                                                   |             |
|--------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------|-------------|
| Processed timestamps (after reindex)       |                                       | 312,048                                                                                                 |             |
| Raw union timestamps                       |                                       | 292,447                                                                                                 |             |
| Synthetic rows (reindexed, forward-filled) |                                       | 19,601                                                                                                  |             |
| Synthetic ratio                            |                                       | 6.28%                                                                                                   |             |
| Example synthetic timestamps               |                                       | 2026-01-08 07:00:00, 2026-01-08 07:10:00, 2026-01-08 07:20:00, 2026-01-08 07:30:00, 2026-01-08 07:40:00 |             |
| Sensor Group                               | Column Pattern                        | Unit                                                                                                    | Valid Range |
| Wind Speed                                 | TB{id}_Ambient WindSpeed Avg.         | m/s                                                                                                     | [0, 60]     |
| Temperature                                | TB{id}_Ambient Temp. Avg.             | deg C                                                                                                   | [-10, 55]   |
| Power (Target)                             | TB{id}_Grid Production Power Avg.     | kW                                                                                                      | [0, 2,200]  |
| Frequency                                  | TB{id}_Grid Production Frequency Avg. | Hz                                                                                                      | [47, 53]    |

## 3.2 Data Quality Analysis

Data quality was assessed by computing missing rates per column. Most turbines show 6.4-7.1% missing data (classified as 'Minor gaps'), while TB05 has the highest turbine missing rate at 10.47% (classified as 'Moderate missing data'). The farm-level aggregate power column has 0.0% missing data; farm-total results below are therefore based on the observed aggregate series, and a missing value in the aggregate is not interpreted as a reliability statement.

| Turbine | Missing Rate (%) | Status                |
|---------|------------------|-----------------------|
| TB01    | 6.49             | Minor gaps            |
| TB02    | 6.44             | Minor gaps            |
| TB03    | 6.87             | Minor gaps            |
| TB04    | 6.43             | Minor gaps            |
| TB05    | 10.47            | Moderate missing data |
| TB06    | 6.40             | Minor gaps            |
| TB07    | 6.55             | Minor gaps            |
| TB08    | 6.41             | Minor gaps            |
| TB09    | 7.08             | Minor gaps            |
| TB10    | 6.61             | Minor gaps            |
| TB11    | 6.48             | Minor gaps            |
| TB12    | 6.92             | Minor gaps            |

## 3.3 Turbine Availability

Availability is computed from the test period (test window ends 2026-07-31). To make the definition unambiguous, three complementary metrics are reported per turbine:

| Metric                            | Formula                                                                                                                       | Meaning                                                                                                                   |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Observed operational availability | $\text{Generating} / (\text{Generating} + \text{Stopped} + \text{Curtailed} + \text{Standby} + \text{Comm. loss}) \times 100$ | Availability over the observed time only (missing telemetry excluded from the denominator).                               |
| Coverage-adjusted availability    | $\text{Generating} / \text{Total elapsed hours} \times 100$                                                                   | Conservative metric that treats missing time as unavailable (missing included in the denominator).                        |
| Data coverage                     | $\text{Observed hours} / \text{Total elapsed hours} \times 100$                                                               | Share of the calendar period with telemetry present (observed = generating + stopped + curtailed + standby + comm. loss). |

Observed operational availability is the headline figure used throughout this report; coverage-adjusted availability and data coverage are reported alongside it so that a low headline value can be attributed to either low generation or missing telemetry (e.g. TB12, Section 5.5).

Availability was computed from the test period. The average availability across all turbines is 82.17%. TB01 has the highest at 84.25%, while TB12 has the lowest at 73.29%.



| Turbine | Generating (hrs) | Stopped (hrs) | Missing (hrs) | Observed Operations Availability (%) | Adjusted Availability (%) | D(%) Cov. (%) |
|---------|------------------|---------------|---------------|--------------------------------------|---------------------------|---------------|
| TB01    | 3,225            | 488           | 878           | 84.25                                | 68.53                     | 81.34         |
| TB02    | 3,208            | 502           | 884           | 83.94                                | 68.17                     | 81.21         |
| TB03    | 3,210            | 489           | 878           | 83.86                                | 68.22                     | 81.34         |
| TB04    | 3,221            | 530           | 878           | 84.15                                | 68.45                     | 81.35         |
| TB05    | 3,131            | 458           | 936           | 83.07                                | 66.55                     | 80.11         |
| TB06    | 3,156            | 629           | 878           | 82.45                                | 67.06                     | 81.34         |
| TB07    | 3,166            | 518           | 933           | 83.94                                | 67.30                     | 80.17         |
| TB08    | 3,128            | 636           | 880           | 81.75                                | 66.47                     | 81.31         |
| TB09    | 3,105            | 613           | 878           | 81.12                                | 65.99                     | 81.35         |
| TB10    | 3,122            | 586           | 933           | 82.77                                | 66.36                     | 80.17         |
| TB11    | 3,119            | 626           | 878           | 81.50                                | 66.29                     | 81.34         |
| TB12    | 2,695            | 627           | 1,028         | 73.29                                | 57.28                     | 78.15         |

## 4. Methodology

### 4.1 Pipeline Architecture

The system implements a 15-step automated pipeline orchestrated by [main.py](#). Each step is modular, with dedicated source files in the [src/](#) directory. The pipeline handles data loading, preprocessing, feature engineering, model training, evaluation, and output generation.

| Step | Module                   | Description                                      |
|------|--------------------------|--------------------------------------------------|
| 1    | load_data.py             | Load 11 raw SCADA Excel files                    |
| 2    | column_mapping.py        | Standardize SCADA column names                   |
| 3    | data_validation.py       | Validate ranges, units, completeness             |
| 3b   | data_validation.py       | Timestamp coverage audit, duplicates, timezone   |
| 4    | preprocessing.py         | Clean, resample, handle missing values           |
| 5    | feature_engineering.py   | Create lags, rolling stats, temporal features    |
| 6    | split_time_series.py     | Time-based 70/15/15 train/val/test split         |
| 7    | train_baseline.py        | Train persistence + linear regression baselines  |
| 8    | train_power_model.py     | Train XGBoost + LightGBM for all targets         |
| 9    | train_anomaly_model.py   | Isolation forest anomaly detection               |
| 10   | train_failure_model.py   | Failure risk analysis + availability             |
| 11   | predict.py + evaluate.py | Generate predictions + compute metrics           |
| 12   | predict.py               | Create forecast output with confidence intervals |
| 13   | evaluate.py              | Generate 6 summary visualizations                |
| 14   | generate_outputs.py      | Generate all doc Section 15 output files         |
| 15   | inventory.py             | Provenance — reindex additions + auto inventory  |

### 4.2 Feature Engineering

Feature engineering creates approximately 150 features per target variable from raw SCADA measurements. The feature categories include:

| Category           | Features                                     | Count         |
|--------------------|----------------------------------------------|---------------|
| Lag Features       | 1, 2, 3, 6, 12, 144 steps                    | 6 per sensor  |
| Rolling Statistics | Mean, Std, Min, Max over [6, 18, 36, 144]    | 16 per sensor |
| Temporal           | Hour, Day of week, Month, Season, Is weekend | 5             |
| Power Change       | Diff [1,3,6], Pct change [1]                 | 4             |
| Ramp Features      | Rolling ramp rate, ramp indicators           | 4             |

| Category         | Features                           | Count |
|------------------|------------------------------------|-------|
| Interactions     | Wind x temperature, power per wind | 4+    |
| Total per target | -                                  | ~150  |

### 4.3 Time Series Split & Validation

Data is split chronologically (70% train / 15% validation / 15% test) to prevent look-ahead bias. The split is strictly time-based; no random shuffle is applied. All feature engineering uses only data available up to the forecast issue time (shift()) for lags, rolling windows computed on past data only). Scaler, imputer, and feature selectors are fit exclusively on the training set.

#### *Train / Validation / Test Split (row provenance):*

Provenance per row is set by the pipeline (is\_observed / is\_synthetic / is\_imputed, Section 3.1): “Observed” = timestamp present in the raw union; “Synthetic” = row inserted by the 10-min reindex; “Imputed” = row whose value was forward-filled. Only rows that are observed and not imputed are eligible to back the official metrics.

| Split            | Rows    | Observed | Synthetic | Imputed | Obs. not imputed | Date Range               |
|------------------|---------|----------|-----------|---------|------------------|--------------------------|
| Train (70%)      | 218,433 | 218,433  | 0         | 2,395   | 216,038          | 2021-01-01 to 2025-02-25 |
| Validation (15%) | 46,807  | 46,087   | 720       | 679     | 45,420           | 2025-02-25 to 2026-01-16 |
| Test (15%)       | 28,232  | 22,887   | 5,345     | 501     | 22,465           | 2026-01-16 to 2026-07-31 |

#### *Valid targets per horizon (honest scoring denominators):*

| Split      | Horizon | Rows    | Valid Targets | Valid Ratio |
|------------|---------|---------|---------------|-------------|
| Train      | 10min   | 218,433 | 217,879       | 99.75%      |
| Train      | 30min   | 218,433 | 217,879       | 99.75%      |
| Train      | 1hour   | 218,433 | 217,879       | 99.75%      |
| Train      | 6hour   | 218,433 | 217,879       | 99.75%      |
| Train      | 24hour  | 218,433 | 217,879       | 99.75%      |
| Validation | 10min   | 46,807  | 45,865        | 97.99%      |
| Validation | 30min   | 46,807  | 45,865        | 97.99%      |
| Validation | 1hour   | 46,807  | 45,865        | 97.99%      |
| Validation | 6hour   | 46,807  | 45,865        | 97.99%      |
| Validation | 24hour  | 46,807  | 45,865        | 97.99%      |
| Test       | 10min   | 28,232  | 22,964        | 81.34%      |
| Test       | 30min   | 28,232  | 22,964        | 81.34%      |
| Test       | 1hour   | 28,232  | 22,964        | 81.34%      |
| Test       | 6hour   | 28,232  | 22,964        | 81.34%      |
| Test       | 24hour  | 28,232  | 22,964        | 81.34%      |

### Data Leakage Prevention:

- Missing value imputation uses forward-fill (no future data in fill)
- Rolling statistics use only past observations (no center=True)
- StandardScaler fit on training data only, transforms on val/test
- Target shift correctly:  $X(t) \rightarrow P(t+h)$ ; no power at  $t$  used as a feature for  $t$
- Lag features use shift(1) so only past observations are consumed
- All splits are chronological; no random shuffle applied
- Feature selection (SelectKBest) fit on training data only

### Walk-Forward Validation:

To assess model stability, walk-forward validation with 5 chronological folds is performed on baseline models (Persistence and Ridge regression). Each fold uses an expanding training window with a held-out test segment. Mean and standard deviation of RMSE and  $R^2$  across folds are reported to quantify performance variability. The 5 folds are the count actually produced by the expanding-window split (walk\_forward\_split), not a requested target.

## 4.4 Model Training

Two gradient boosting algorithms were used: XGBoost and LightGBM. Both models were trained with identical hyperparameters for fair comparison. Each target variable (12 turbines + farm aggregate x 5 horizons) received independent model training, resulting in **130 ML models** (13 targets x 2 algorithms x 5 horizons). This count is read from the actual \*\_model.joblib files in models/ via inventory\_summary.json, not computed by multiplication. Each trained key is stored as 4 artifacts (model + scaler + feature list + metadata), i.e. **520 files**; the 10 baseline evaluations (persistence + ridge x 5 horizons) are trained and evaluated separately via walk-forward validation.

| Hyperparameter   | XGBoost | LightGBM |
|------------------|---------|----------|
| n_estimators     | 150     | 150      |
| max_depth        | 6       | 6        |
| learning_rate    | 0.1     | 0.1      |
| subsample        | 0.8     | 0.8      |
| colsample_bytree | 0.5     | 0.5      |
| random_state     | 42      | 42       |

**Hyperparameter tuning (evidence trail):** the pipeline exposes an optional tuning stage ([training.tuning.enabled](#) in config.yaml, default **false**). When enabled, it runs a 12-iteration Bayesian-style random search on XGBoost and LightGBM using TimeSeriesSplit cross-validation (3 folds, expanding window) per turbine-horizon target, records the per-fold and mean CV RMSE for every trial, and persists two evidence files: [data/metadata/tuning\\_results.csv](#) (per-trial, per-fold, mean CV RMSE) and [data/metadata/best\\_params.json](#) (winning hyperparameters per target). The shipped models in this report use the fixed hyperparameters in the table above (tuning disabled); re-running [main.py](#) with tuning enabled replaces those parameters and records the full evidence trail for re-audit.

### Leakage Audit (every trained model):

The pipeline runs an automated leakage audit over every trained model. Each model's persisted feature list is checked for any target/future marker (e.g. '\_target\_', '\_missing\_', '\_status\_', '\_is\_anomaly\_', '\_anomaly\_score\_', '\_is\_stopped\_', '\_failure\_event\_'); the audit also verifies that the target column  $P(t+h)$  is not among the features, that no future-leaning feature is present, that timestamps satisfy  $\text{timestamp\_target} = \text{timestamp\_issue} + \text{horizon}$ , and that no two models return predictions identical to their target. The audit covers **180 model fits** across ridge, XGBoost and LightGBM (120 ML fits), each checked per turbine and per horizon (10 min/30 min/1 h/6 h/24 h). Breakdown: ridge: 60 models, 0 flagged; xgboost: 60 models, 0 flagged; lightgbm: 60 models, 0 flagged. Result (full audit, every turbine x horizon x model family): **0 flagged models out of 180** audited (55,800 features used in total). A non-zero count aborts the pipeline (fail-closed); the per-family audit (leakage\_audit\_full.csv) is persisted to [data/metadata/leakage\\_audit\\_full.csv](#).

## 4.5 Evaluation Metrics

| Metric      | Formula                                                        | Description                                                |
|-------------|----------------------------------------------------------------|------------------------------------------------------------|
| MAE         | $\text{Mean}( F - O )$                                         | Mean Absolute Error (kW)                                   |
| nMAE        | $\text{MAE} / P_{\text{rated}} \times 100\%$                   | Normalized MAE (%)                                         |
| RMSE        | $\sqrt{\text{Mean}((F - O)^2)}$                                | Root Mean Square Error (kW)                                |
| nRMSE       | $\text{RMSE} / P_{\text{rated}} \times 100\%$                  | Normalized RMSE (%)                                        |
| Bias        | $\text{Mean}(F - O)$                                           | Mean Error (kW), detects over/under-forecast               |
| R2          | $1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$            | Coefficient of Determination                               |
| Skill Score | $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{baseline}}$ | Improvement over a stated baseline (0 = same, 1 = perfect) |
| n_samples   | $\text{count}(\text{valid } P(t+h))$                           | Explicit scoring denominator per horizon                   |
| Max Error   | $\max( F - O )$                                                | Worst-case error (kW)                                      |

Per P0-03 review comment, every skill score is reported with its **explicit baseline** and its **sample size**: 'Skill vs persistence' =  $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{persistence}}$  and 'Skill vs Ridge' =  $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{ridge}}$ , both computed on the **identical set of test samples** where all three models (boosting, persistence, Ridge) have a valid target  $P(t+h)$ . n\_samples is stored per row in [evaluation\\_metrics.csv](#).

## 5. Results

### 5.1 Model Performance Overview

The following table summarizes mean turbine R2 (average of 12 individual turbine R2 values, from *metrics.csv*) per model per horizon. Ridge is the champion (min mean RMSE) in 4 of 5 turbine cells. LightGBM is the champion in 1 of 5 turbine cells (6hour). At turbine level, LightGBM and XGBoost mean  $R^2$  differ by at most 0.014 across horizons. Best mean turbine R2 degrades from 0.942 at 10-min to 0.344 at 24-hour, consistent with the inherent difficulty of longer-range wind power prediction.

| Horizon | Model       | MAE   | nMAE% | RMSE  | nRMSE% | Bias  | R2      | SkIP  | SkIR   | n      |
|---------|-------------|-------|-------|-------|--------|-------|---------|-------|--------|--------|
| 10min   | lightgbm    | 98.2  | 4.5   | 154.2 | 7.0    | +5.0  | 0.9338  | 0.012 | -0.055 | 22,306 |
| 10min   | persistence | 92.0  | 4.2   | 156.2 | 7.1    | +0.2  | 0.9336  | 0.000 | -0.069 | 22,304 |
| 10min   | ridge       | 91.5  | 4.2   | 146.2 | 6.6    | -1.2  | 0.9419  | 0.064 | 0.000  | 22,306 |
| 10min   | xgboost     | 99.7  | 4.5   | 155.3 | 7.1    | +6.0  | 0.9330  | 0.005 | -0.062 | 22,306 |
| 1hour   | lightgbm    | 224.1 | 10.2  | 316.1 | 14.4   | +4.5  | 0.7278  | 0.082 | -0.033 | 22,311 |
| 1hour   | persistence | 213.6 | 9.7   | 344.5 | 15.7   | +1.2  | 0.6778  | 0.000 | -0.126 | 22,304 |
| 1hour   | ridge       | 211.6 | 9.6   | 306.1 | 13.9   | -3.4  | 0.7453  | 0.111 | 0.000  | 22,311 |
| 1hour   | xgboost     | 230.8 | 10.5  | 320.0 | 14.5   | +12.7 | 0.7208  | 0.071 | -0.046 | 22,311 |
| 24hour  | lightgbm    | 405.1 | 18.4  | 511.2 | 23.2   | +15.5 | 0.2883  | 0.129 | -0.043 | 21,415 |
| 24hour  | persistence | 396.2 | 18.0  | 585.5 | 26.6   | -7.7  | 0.0650  | 0.000 | -0.198 | 21,315 |
| 24hour  | ridge       | 378.6 | 17.2  | 490.5 | 22.3   | -6.0  | 0.3443  | 0.164 | 0.000  | 21,415 |
| 24hour  | xgboost     | 412.8 | 18.8  | 516.2 | 23.5   | +29.0 | 0.2742  | 0.120 | -0.054 | 21,415 |
| 30min   | lightgbm    | 167.3 | 7.6   | 246.3 | 11.2   | +8.1  | 0.8344  | 0.055 | -0.029 | 22,308 |
| 30min   | persistence | 156.8 | 7.1   | 260.6 | 11.8   | +0.6  | 0.8155  | 0.000 | -0.089 | 22,304 |
| 30min   | ridge       | 158.0 | 7.2   | 239.4 | 10.9   | -2.3  | 0.8442  | 0.081 | 0.000  | 22,308 |
| 30min   | xgboost     | 173.1 | 7.9   | 250.5 | 11.4   | +13.6 | 0.8284  | 0.038 | -0.047 | 22,308 |
| 6hour   | lightgbm    | 349.4 | 15.9  | 453.7 | 20.6   | +10.8 | 0.4374  | 0.371 | 0.023  | 22,159 |
| 6hour   | persistence | 518.5 | 23.6  | 723.5 | 32.9   | -1.6  | -0.4257 | 0.000 | -0.560 | 22,124 |
| 6hour   | ridge       | 358.8 | 16.3  | 464.9 | 21.1   | +9.3  | 0.4094  | 0.356 | 0.000  | 22,159 |
| 6hour   | xgboost     | 355.6 | 16.2  | 457.8 | 20.8   | +19.1 | 0.4276  | 0.366 | 0.015  | 22,159 |

Mean of the 12 turbine rows per (horizon, model) cell of *metrics.csv*. SkIP = skill vs persistence; SkIR = skill vs Ridge; both on identical samples (Section 4.5). n = mean valid test samples per row.

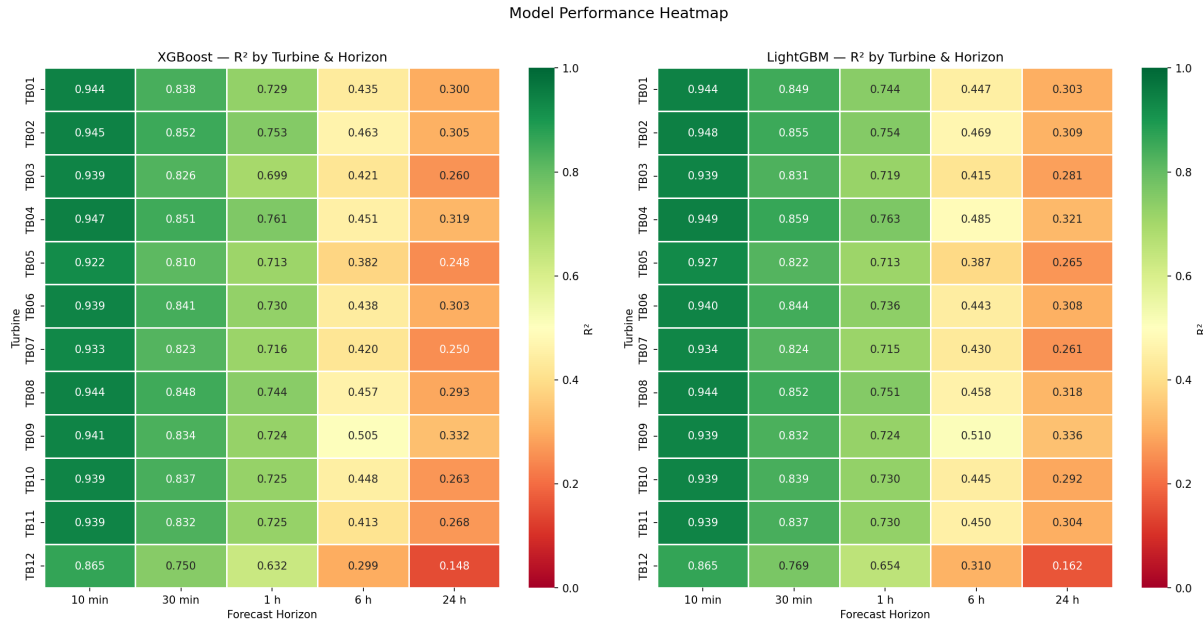


Figure 1: Model performance heatmap across turbines and horizons

### Walk-Forward Validation (Baselines):

Walk-forward validation across 5 chronological folds (baselines). Values are mean +/- standard deviation across folds.

| Model       | Horizon | RMSE (kW)         | R2                 |
|-------------|---------|-------------------|--------------------|
| Persistence | 10min   | 267.4 +/- 333.4   | 0.9253 +/- 0.0134  |
| Persistence | 30min   | 472.9 +/- 651.7   | 0.7904 +/- 0.0276  |
| Persistence | 1hour   | 629.1 +/- 902.5   | 0.6466 +/- 0.0370  |
| Persistence | 6hour   | 1224.3 +/- 1871.0 | -0.2346 +/- 0.1078 |
| Persistence | 24hour  | 1187.8 +/- 1811.2 | -0.1642 +/- 0.0883 |
| Ridge       | 10min   | 263.3 +/- 318.0   | 0.9223 +/- 0.0473  |
| Ridge       | 30min   | 449.4 +/- 620.8   | 0.8108 +/- 0.0331  |
| Ridge       | 1hour   | 583.2 +/- 833.0   | 0.6921 +/- 0.0774  |
| Ridge       | 6hour   | 914.9 +/- 1387.6  | 0.2988 +/- 0.1280  |
| Ridge       | 24hour  | 1023.8 +/- 1577.7 | 0.1432 +/- 0.1094  |

**Key evidence:** Persistence 10min: RMSE=267.4+/-333.4 kW, R2=0.9253+/-0.0134; Persistence 1hour: RMSE=629.1+/-902.5 kW, R2=0.6466+/-0.0370; Persistence 24hour: RMSE=1187.8+/-1811.2 kW, R2=-0.1642+/-0.0883; Persistence 30min: RMSE=472.9+/-651.7 kW, R2=0.7904+/-0.0276.

## 5.2 Horizon Decay Analysis

As expected, forecast accuracy decreases with increasing horizon. The R<sup>2</sup> decay is approximately linear from 10-min to 1-hour, then accelerates for 6-hour and 24-hour horizons. This is consistent with the inherent difficulty of longer-range wind power prediction.

**Evidence:** 10min lightgbm:  $R^2=0.9338\pm0.0226$ ; 10min persistence:  $R^2=0.9336\pm0.0054$ ; 10min ridge:  $R^2=0.9419\pm0.0052$ ; 10min xgboost:  $R^2=0.9330\pm0.0223$ ; 1hour lightgbm:  $R^2=0.7278\pm0.0281$ ; 1hour persistence:  $R^2=0.6778\pm0.0178$ .

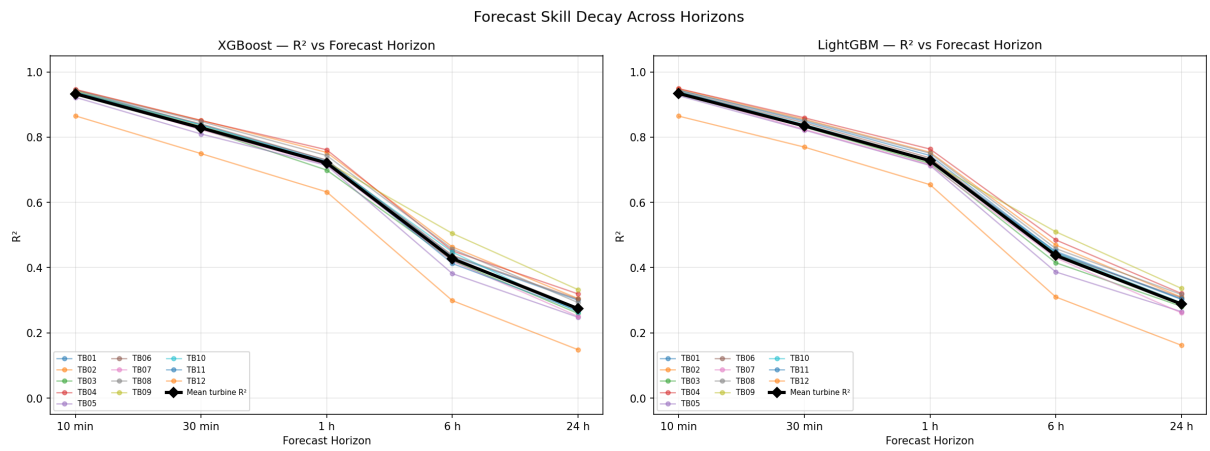


Figure 2: R2 degradation across forecast horizons



### 5.3 Model Comparison (turbine-avg)

XGBoost vs LightGBM mean turbine  $R^2$  gap per horizon: 10min:  $|\Delta R^2|=0.0008$ ; 1hour:  $|\Delta R^2|=0.0070$ ; 24hour:  $|\Delta R^2|=0.0141$ ; 30min:  $|\Delta R^2|=0.0060$ ; 6hour:  $|\Delta R^2|=0.0098$ .

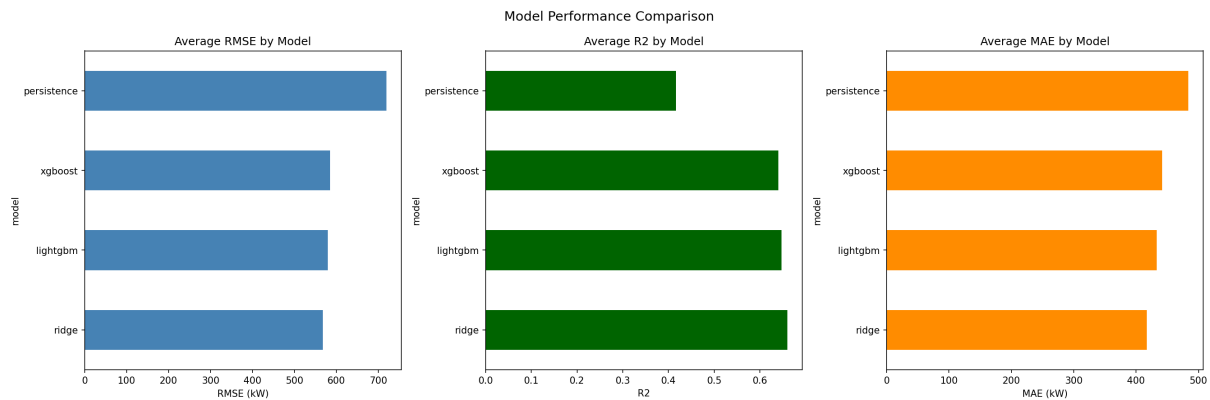


Figure 3: XGBoost vs LightGBM comparison by horizon

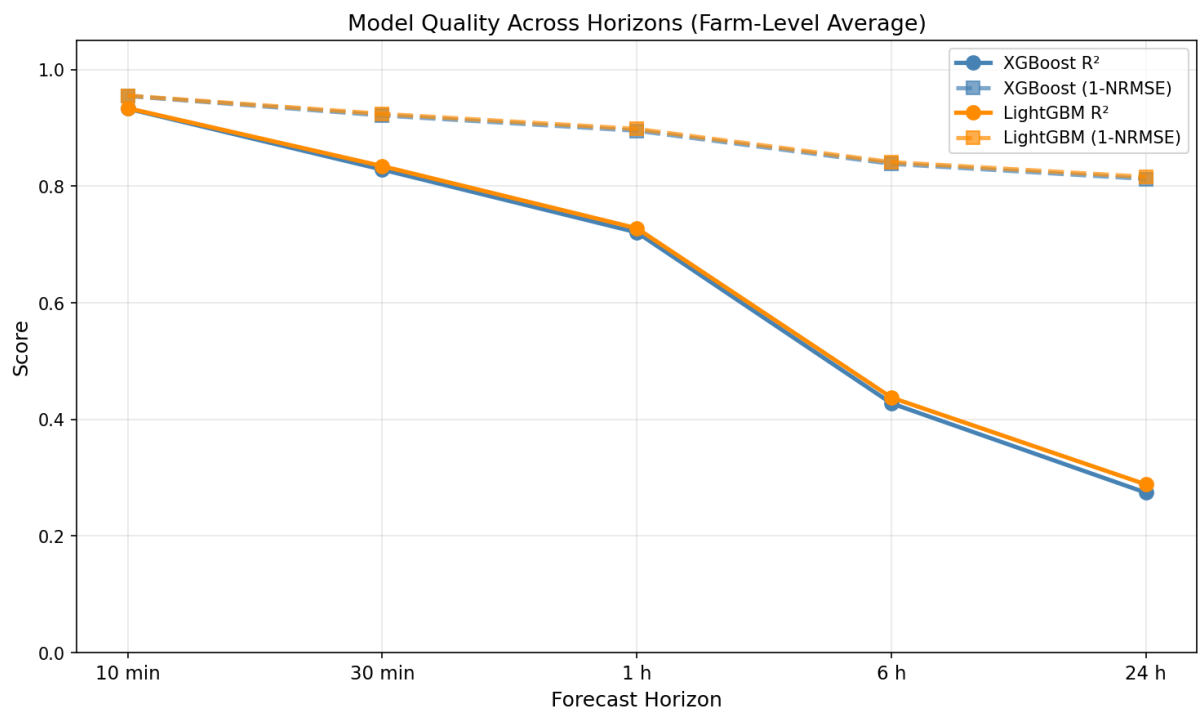


Figure 4: Multi-metric radar comparison of model performance

### 5.4 Farm-Level Results

Farm-level metrics are calculated **directly** on the summed farm total power (actual and predicted), not as an average of individual turbine  $R^2$ . This gives a proper assessment of farm-level forecast accuracy.

| Horizon | Model    | MAE (kW) | RMSE (kW) | nRMSE% | Bias   | R2     | R2 corr. | Bias corr. |
|---------|----------|----------|-----------|--------|--------|--------|----------|------------|
| 10min   | xgboost  | 800.6    | 1209.7    | 4.6    | +106.1 | 0.9691 | 0.9693   | +6.7       |
| 10min   | lightgbm | 794.0    | 1216.2    | 4.6    | +104.7 | 0.9688 | 0.9688   | +39.1      |

| Horizon | Model    | MAE (kW) | RMSE (kW) | nRMSE% | Bias   | R2     | R2 corr. | Bias corr. |
|---------|----------|----------|-----------|--------|--------|--------|----------|------------|
| 30min   | xgboost  | 1728.3   | 2453.9    | 9.3    | +268.0 | 0.8728 | 0.8748   | -83.9      |
| 30min   | lightgbm | 1591.0   | 2347.7    | 8.9    | +130.5 | 0.8836 | 0.8840   | +45.1      |
| 1hour   | xgboost  | 2342.0   | 3231.1    | 12.2   | +186.0 | 0.7794 | 0.7828   | -18.2      |
| 1hour   | lightgbm | 2257.7   | 3198.7    | 12.1   | +77.0  | 0.7838 | 0.7856   | +87.5      |
| 6hour   | xgboost  | 3918.7   | 4962.3    | 18.8   | +350.5 | 0.4762 | 0.4836   | -129.4     |
| 6hour   | lightgbm | 3873.4   | 4945.6    | 18.7   | +198.4 | 0.4797 | 0.4843   | -63.5      |
| 24hour  | xgboost  | 4669.0   | 5791.9    | 21.9   | +560.2 | 0.2863 | 0.2821   | +237.2     |
| 24hour  | lightgbm | 4682.8   | 5789.1    | 21.9   | +545.7 | 0.2870 | 0.2828   | +179.5     |

**Evidence:** Farm-level  $R^2$  ranges from 0.9691 (10min). The 10-min farm forecast achieves  $R^2=0.9691$  with MAE=801 kW (nMAE=3.0%).

**Farm-total champions per horizon** (min mean RMSE from *farm\_metrics.csv*): 10min: XGBoost RMSE 1210 kW ( $R^2=0.9691$ ); 1hour: LightGBM RMSE 3199 kW ( $R^2=0.7838$ ); 24hour: LightGBM RMSE 5789 kW ( $R^2=0.2870$ ); 30min: LightGBM RMSE 2348 kW ( $R^2=0.8836$ ); 6hour: LightGBM RMSE 4946 kW ( $R^2=0.4797$ ).

At farm-total level, LightGBM and XGBoost  $R^2$  differ by at most 0.011 across horizons (from *farm\_metrics.csv*).

**Bias correction (P1-04):** a per-model linear offset fitted on the validation split (corrected = slope  $\times$  predicted + intercept) was applied to the test forecasts. Best corrected  $R^2=0.9693$  (10min, xgboost); raw/corrected columns are shown side by side above and every corrected metric is scored on the same samples as the raw one.

#### 5.4.1 Farm Bias Analysis (P1-04)

Model bias is examined by comparing the direct farm-total forecast with the SUM of the 12 individual turbine forecasts, and against the observed farm power ( $P(t+h)$ ). A positive bias means the farm model over-forecasts on average. This directly addresses the review concern that reported skill may hide systematic under-/over-forecasting.

| Horizon | n      | Actual (kW) | Farm model (kW) | Bias (kW) | Bias % rated | MAE (kW) | Farm vs sum (kW) |
|---------|--------|-------------|-----------------|-----------|--------------|----------|------------------|
| 10min   | 28,232 | 5,536       | 5,783           | +247.0    | +0.936       | 813.8    | -59.5            |
| 30min   | 28,232 | 5,537       | 6,644           | +1,107.0  | +4.193       | 2,299.4  | +464.0           |
| 1hour   | 28,232 | 5,538       | 6,229           | +690.8    | +2.617       | 2,459.6  | +61.5            |
| 6hour   | 28,232 | 5,549       | 6,998           | +1,448.7  | +5.488       | 4,393.7  | -102.9           |
| 24hour  | 28,232 | 5,598       | 6,569           | +970.7    | +3.677       | 4,670.3  | -504.3           |

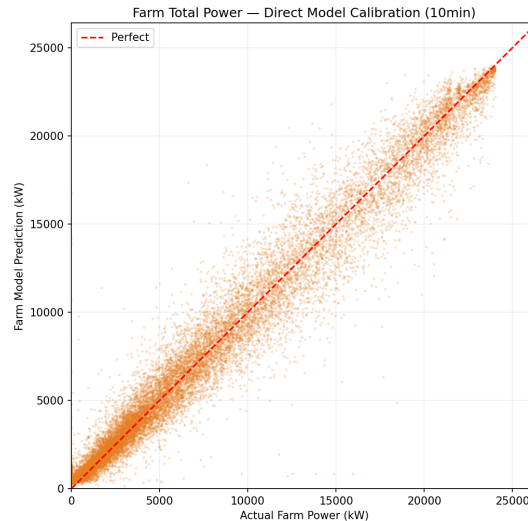


Figure 5: Farm-level bias vs predicted power with calibration reference (identity line)

#### 5.4.2 Horizon Comparison Window Check (P1-04)

To rule out a sampling artifact,  $R^2$  for every horizon pair is recomputed on the **intersection** of valid target samples — i.e. the same test window (same timestamps) and the same `n_at_capacity` / `n_zero_power` masks. If the 24h > 6h  $R^2$  result survives on identical samples, it is a genuine forecast-quality effect rather than a target-shift artifact.

| Horizon A | Horizon B | n common | Window same | R2 A (common) | R2 B (common) | Delta R2 |
|-----------|-----------|----------|-------------|---------------|---------------|----------|
| 10min     | 24hour    | 28,232   | yes         | 0.9702        | 0.2848        | -0.6853  |
| 30min     | 24hour    | 28,232   | yes         | 0.7966        | 0.2848        | -0.5117  |
| 1hour     | 24hour    | 28,232   | yes         | 0.7752        | 0.2848        | -0.4904  |
| 6hour     | 24hour    | 28,232   | yes         | 0.3656        | 0.2848        | -0.0808  |

### 5.5 TB12 Turbine Analysis

Turbine TB12 shows significantly lower forecast accuracy compared to other turbines. In the official test window (28,232 rows) it has 21.85% missing data, 20.87% stopped/near-zero power output, and a 19.84% frozen-data ratio (225 blocks). Across the full processed dataset TB12\_power is 6.92% missing after forward-fill (per-column, Section 3.2); the test-window rate reflects the evaluation period only. Its wind-power correlation within the official window is 0.8402. These data-quality issues warrant investigation into sensor calibration and operating conditions.

| Metric                        | Value |
|-------------------------------|-------|
| Missing Data Rate (%)         | 21.85 |
| Stopped / Near-Zero Power (%) | 20.87 |
| Frozen/Stuck Data Ratio (%)   | 19.84 |
| Frozen Data Blocks            | 225   |
| Mean Power (kW)               | 461.7 |

| Metric                         | Value   |
|--------------------------------|---------|
| Std Power (kW, when operating) | 564.6   |
| Mean Power TB09 (kW)           | 553.8   |
| Mean Power TB04 (kW)           | 620.4   |
| Power Correlation with TB09    | 0.7647  |
| Power Correlation with TB04    | 0.7905  |
| Mean Wind Speed TB09 (m/s)     | 5.84    |
| Wind-Power Correlation TB12    | 0.8402  |
|                                |         |
| R2 10min (lightgbm)            | 0.8645  |
| R2 10min (persistence)         | 0.9267  |
| R2 10min (ridge)               | 0.931   |
| R2 10min (xgboost)             | 0.8653  |
| R2 1hour (lightgbm)            | 0.6543  |
| R2 1hour (persistence)         | 0.6643  |
| R2 1hour (ridge)               | 0.7219  |
| R2 1hour (xgboost)             | 0.6321  |
| R2 24hour (lightgbm)           | 0.1619  |
| R2 24hour (persistence)        | -0.1313 |
| R2 24hour (ridge)              | 0.2193  |
| R2 24hour (xgboost)            | 0.1483  |
| R2 30min (lightgbm)            | 0.7695  |
| R2 30min (persistence)         | 0.8021  |
| R2 30min (ridge)               | 0.8247  |
| R2 30min (xgboost)             | 0.7496  |
| R2 6hour (lightgbm)            | 0.3101  |
| R2 6hour (persistence)         | -0.4069 |
| R2 6hour (ridge)               | 0.3043  |
| R2 6hour (xgboost)             | 0.2988  |

TB12 shows higher missing data rate and longer stop periods compared to TB09 and TB04. Its power-wind correlation is weaker, suggesting possible sensor drift or wake effects. A dedicated sensor recalibration and data quality investigation is recommended.

## 5.6 Operational Analysis

Error analysis by operating regime provides insight into model behavior under different conditions. The model comparison and radar summary above show consistent performance characteristics across turbines.

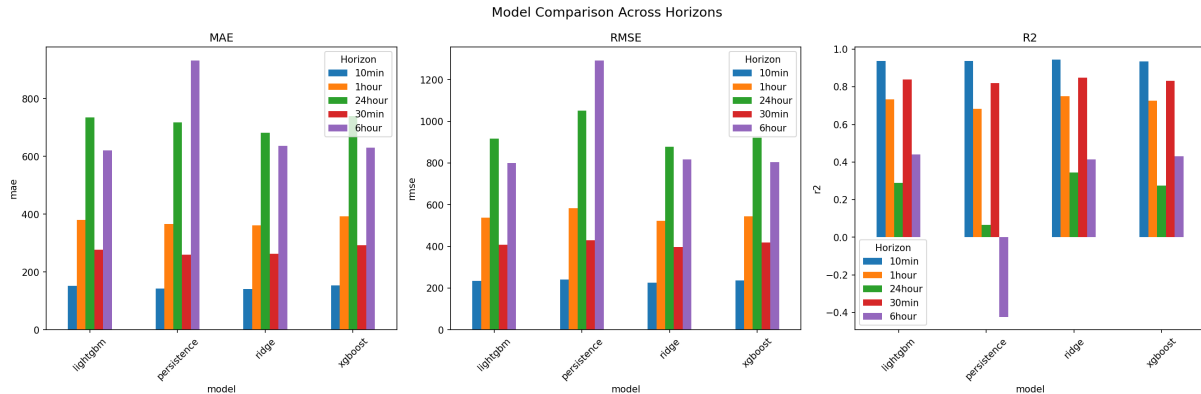


Figure 6: Horizon-wise model performance comparison

## 5.7 Alert Accuracy

Ramp event detection accuracy is evaluated using Precision, Recall, F1-score, and False Alarm Rate (FAR). A ramp event is defined as a power change exceeding 0.5% per minute of rated capacity.

**Alert semantics:** all ramp, anomaly, temperature and failure-risk alerts are **informational advisories** produced by heuristic screening (`method=heuristic_screening`, `confirmed=False`, `verification_status=SCREENING_ONLY` in every alert row and in `data/metadata/alert_screening_summary.json`). They do not replace operator decisions, SCADA trip logic, or safety interlocks, and they never modify generation set-points. Accuracy metrics in this section therefore measure the detection quality of the advisory stream against the heuristic ground-truth rules, not an operational control function or confirmed event forecasts.

**Evidence:** Average Precision=0.805, Recall=0.855, F1=0.829 across all turbine-horizon-model combinations.

| Model                | Precision | Recall | F1    | FAR   | TP    | FP   | FN   |
|----------------------|-----------|--------|-------|-------|-------|------|------|
| xgboost_TB01_10min   | 0.883     | 0.910  | 0.897 | 0.117 | 15569 | 2053 | 1530 |
| lightgbm_TB01_10min  | 0.897     | 0.915  | 0.906 | 0.102 | 15654 | 1788 | 1445 |
| xgboost_TB01_30min   | 0.816     | 0.890  | 0.851 | 0.184 | 15217 | 3443 | 1882 |
| lightgbm_TB01_30min  | 0.850     | 0.892  | 0.871 | 0.150 | 15259 | 2696 | 1840 |
| xgboost_TB01_1hour   | 0.812     | 0.883  | 0.846 | 0.188 | 15091 | 3499 | 2008 |
| lightgbm_TB01_1hour  | 0.838     | 0.878  | 0.857 | 0.162 | 15009 | 2901 | 2090 |
| xgboost_TB01_6hour   | 0.744     | 0.814  | 0.777 | 0.256 | 13924 | 4793 | 3175 |
| lightgbm_TB01_6hour  | 0.770     | 0.802  | 0.786 | 0.230 | 13720 | 4090 | 3379 |
| xgboost_TB01_24hour  | 0.756     | 0.829  | 0.791 | 0.244 | 14176 | 4579 | 2923 |
| lightgbm_TB01_24hour | 0.772     | 0.801  | 0.786 | 0.228 | 13698 | 4052 | 3401 |

| Model                | Precision | Recall | F1    | FAR   | TP    | FP   | FN   |
|----------------------|-----------|--------|-------|-------|-------|------|------|
| xgboost_TB02_10min   | 0.872     | 0.910  | 0.890 | 0.128 | 15446 | 2269 | 1535 |
| lightgbm_TB02_10min  | 0.900     | 0.909  | 0.905 | 0.100 | 15440 | 1715 | 1541 |
| xgboost_TB02_30min   | 0.854     | 0.878  | 0.866 | 0.146 | 14904 | 2547 | 2077 |
| lightgbm_TB02_30min  | 0.858     | 0.877  | 0.868 | 0.142 | 14897 | 2458 | 2084 |
| xgboost_TB02_1hour   | 0.846     | 0.871  | 0.858 | 0.154 | 14791 | 2703 | 2190 |
| lightgbm_TB02_1hour  | 0.850     | 0.870  | 0.860 | 0.150 | 14773 | 2617 | 2208 |
| xgboost_TB02_6hour   | 0.745     | 0.816  | 0.779 | 0.255 | 13851 | 4732 | 3130 |
| lightgbm_TB02_6hour  | 0.769     | 0.789  | 0.779 | 0.231 | 13399 | 4031 | 3582 |
| xgboost_TB02_24hour  | 0.754     | 0.830  | 0.790 | 0.245 | 14085 | 4582 | 2896 |
| lightgbm_TB02_24hour | 0.770     | 0.795  | 0.782 | 0.230 | 13507 | 4035 | 3474 |
| xgboost_TB03_10min   | 0.877     | 0.921  | 0.898 | 0.123 | 15821 | 2221 | 1363 |
| lightgbm_TB03_10min  | 0.894     | 0.915  | 0.905 | 0.106 | 15727 | 1859 | 1457 |
| xgboost_TB03_30min   | 0.842     | 0.888  | 0.864 | 0.158 | 15258 | 2867 | 1926 |
| lightgbm_TB03_30min  | 0.875     | 0.889  | 0.882 | 0.125 | 15275 | 2179 | 1909 |
| xgboost_TB03_1hour   | 0.796     | 0.882  | 0.837 | 0.204 | 15162 | 3876 | 2022 |
| lightgbm_TB03_1hour  | 0.829     | 0.878  | 0.853 | 0.171 | 15087 | 3111 | 2097 |
| xgboost_TB03_6hour   | 0.767     | 0.806  | 0.786 | 0.233 | 13849 | 4195 | 3335 |
| lightgbm_TB03_6hour  | 0.782     | 0.783  | 0.782 | 0.218 | 13450 | 3754 | 3734 |
| xgboost_TB03_24hour  | 0.766     | 0.833  | 0.798 | 0.234 | 14318 | 4383 | 2866 |
| lightgbm_TB03_24hour | 0.778     | 0.803  | 0.790 | 0.222 | 13805 | 3943 | 3379 |
| xgboost_TB04_10min   | 0.882     | 0.914  | 0.898 | 0.118 | 15613 | 2084 | 1469 |
| lightgbm_TB04_10min  | 0.902     | 0.921  | 0.911 | 0.098 | 15725 | 1710 | 1357 |
| xgboost_TB04_30min   | 0.837     | 0.888  | 0.862 | 0.163 | 15175 | 2951 | 1907 |
| lightgbm_TB04_30min  | 0.848     | 0.889  | 0.868 | 0.152 | 15185 | 2716 | 1897 |
| xgboost_TB04_1hour   | 0.818     | 0.879  | 0.848 | 0.182 | 15020 | 3337 | 2062 |
| lightgbm_TB04_1hour  | 0.837     | 0.878  | 0.857 | 0.163 | 15000 | 2912 | 2082 |
| xgboost_TB04_6hour   | 0.753     | 0.827  | 0.788 | 0.247 | 14126 | 4634 | 2956 |
| lightgbm_TB04_6hour  | 0.778     | 0.796  | 0.787 | 0.222 | 13592 | 3868 | 3490 |
| xgboost_TB04_24hour  | 0.762     | 0.850  | 0.803 | 0.238 | 14518 | 4545 | 2564 |
| lightgbm_TB04_24hour | 0.773     | 0.815  | 0.794 | 0.227 | 13928 | 4083 | 3154 |
| xgboost_TB05_10min   | 0.841     | 0.928  | 0.882 | 0.160 | 15306 | 2904 | 1188 |
| lightgbm_TB05_10min  | 0.860     | 0.923  | 0.890 | 0.140 | 15232 | 2483 | 1262 |
| xgboost_TB05_30min   | 0.793     | 0.900  | 0.844 | 0.206 | 14846 | 3863 | 1648 |

| Model                | Precision | Recall | F1    | FAR   | TP    | FP   | FN   |
|----------------------|-----------|--------|-------|-------|-------|------|------|
| lightgbm_TB05_30min  | 0.826     | 0.901  | 0.862 | 0.174 | 14859 | 3142 | 1635 |
| xgboost_TB05_1hour   | 0.807     | 0.887  | 0.845 | 0.193 | 14628 | 3498 | 1866 |
| lightgbm_TB05_1hour  | 0.805     | 0.888  | 0.845 | 0.195 | 14651 | 3539 | 1843 |
| xgboost_TB05_6hour   | 0.722     | 0.820  | 0.768 | 0.278 | 13521 | 5203 | 2973 |
| lightgbm_TB05_6hour  | 0.741     | 0.812  | 0.775 | 0.260 | 13393 | 4694 | 3101 |
| xgboost_TB05_24hour  | 0.733     | 0.838  | 0.782 | 0.267 | 13827 | 5037 | 2667 |
| lightgbm_TB05_24hour | 0.745     | 0.791  | 0.767 | 0.255 | 13047 | 4456 | 3447 |
| xgboost_TB06_10min   | 0.900     | 0.907  | 0.903 | 0.100 | 15340 | 1703 | 1572 |
| lightgbm_TB06_10min  | 0.903     | 0.911  | 0.907 | 0.097 | 15406 | 1652 | 1506 |
| xgboost_TB06_30min   | 0.847     | 0.870  | 0.859 | 0.153 | 14715 | 2652 | 2197 |
| lightgbm_TB06_30min  | 0.848     | 0.870  | 0.859 | 0.152 | 14713 | 2640 | 2199 |
| xgboost_TB06_1hour   | 0.806     | 0.859  | 0.832 | 0.194 | 14524 | 3500 | 2388 |
| lightgbm_TB06_1hour  | 0.841     | 0.859  | 0.850 | 0.159 | 14523 | 2742 | 2389 |
| xgboost_TB06_6hour   | 0.755     | 0.806  | 0.780 | 0.245 | 13635 | 4426 | 3277 |
| lightgbm_TB06_6hour  | 0.770     | 0.771  | 0.771 | 0.230 | 13047 | 3886 | 3865 |
| xgboost_TB06_24hour  | 0.758     | 0.834  | 0.794 | 0.242 | 14096 | 4494 | 2816 |
| lightgbm_TB06_24hour | 0.780     | 0.789  | 0.784 | 0.220 | 13339 | 3765 | 3573 |
| xgboost_TB07_10min   | 0.882     | 0.908  | 0.895 | 0.118 | 15320 | 2054 | 1548 |
| lightgbm_TB07_10min  | 0.895     | 0.916  | 0.905 | 0.105 | 15453 | 1816 | 1415 |
| xgboost_TB07_30min   | 0.834     | 0.876  | 0.855 | 0.166 | 14782 | 2947 | 2086 |
| lightgbm_TB07_30min  | 0.856     | 0.881  | 0.868 | 0.144 | 14855 | 2507 | 2013 |
| xgboost_TB07_1hour   | 0.840     | 0.862  | 0.851 | 0.160 | 14539 | 2764 | 2329 |
| lightgbm_TB07_1hour  | 0.844     | 0.866  | 0.855 | 0.156 | 14599 | 2696 | 2269 |
| xgboost_TB07_6hour   | 0.746     | 0.808  | 0.776 | 0.254 | 13623 | 4628 | 3245 |
| lightgbm_TB07_6hour  | 0.764     | 0.772  | 0.768 | 0.236 | 13018 | 4018 | 3850 |
| xgboost_TB07_24hour  | 0.750     | 0.830  | 0.788 | 0.250 | 14006 | 4679 | 2862 |
| lightgbm_TB07_24hour | 0.765     | 0.801  | 0.783 | 0.235 | 13515 | 4144 | 3353 |
| xgboost_TB08_10min   | 0.901     | 0.903  | 0.902 | 0.099 | 15094 | 1664 | 1621 |
| lightgbm_TB08_10min  | 0.901     | 0.905  | 0.903 | 0.099 | 15133 | 1653 | 1582 |
| xgboost_TB08_30min   | 0.845     | 0.872  | 0.858 | 0.155 | 14571 | 2664 | 2144 |
| lightgbm_TB08_30min  | 0.851     | 0.876  | 0.864 | 0.149 | 14651 | 2560 | 2064 |
| xgboost_TB08_1hour   | 0.804     | 0.874  | 0.838 | 0.196 | 14614 | 3565 | 2101 |
| lightgbm_TB08_1hour  | 0.856     | 0.867  | 0.862 | 0.144 | 14496 | 2438 | 2219 |

| Model                | Precision | Recall | F1    | FAR   | TP    | FP   | FN   |
|----------------------|-----------|--------|-------|-------|-------|------|------|
| xgboost_TB08_6hour   | 0.739     | 0.801  | 0.769 | 0.261 | 13390 | 4726 | 3325 |
| lightgbm_TB08_6hour  | 0.769     | 0.769  | 0.769 | 0.231 | 12855 | 3862 | 3860 |
| xgboost_TB08_24hour  | 0.754     | 0.835  | 0.792 | 0.246 | 13957 | 4563 | 2758 |
| lightgbm_TB08_24hour | 0.768     | 0.806  | 0.787 | 0.232 | 13481 | 4076 | 3234 |
| xgboost_TB09_10min   | 0.869     | 0.911  | 0.889 | 0.131 | 14993 | 2260 | 1465 |
| lightgbm_TB09_10min  | 0.891     | 0.909  | 0.900 | 0.109 | 14965 | 1839 | 1493 |
| xgboost_TB09_30min   | 0.832     | 0.884  | 0.857 | 0.168 | 14551 | 2945 | 1907 |
| lightgbm_TB09_30min  | 0.844     | 0.882  | 0.863 | 0.156 | 14524 | 2680 | 1934 |
| xgboost_TB09_1hour   | 0.825     | 0.876  | 0.850 | 0.175 | 14414 | 3063 | 2044 |
| lightgbm_TB09_1hour  | 0.832     | 0.873  | 0.852 | 0.168 | 14374 | 2899 | 2084 |
| xgboost_TB09_6hour   | 0.728     | 0.773  | 0.750 | 0.272 | 12724 | 4764 | 3734 |
| lightgbm_TB09_6hour  | 0.748     | 0.757  | 0.753 | 0.252 | 12461 | 4195 | 3997 |
| xgboost_TB09_24hour  | 0.739     | 0.845  | 0.788 | 0.262 | 13912 | 4926 | 2546 |
| lightgbm_TB09_24hour | 0.753     | 0.820  | 0.785 | 0.247 | 13499 | 4424 | 2959 |
| xgboost_TB10_10min   | 0.895     | 0.912  | 0.904 | 0.104 | 15435 | 1802 | 1485 |
| lightgbm_TB10_10min  | 0.902     | 0.908  | 0.905 | 0.098 | 15367 | 1664 | 1553 |
| xgboost_TB10_30min   | 0.855     | 0.884  | 0.869 | 0.145 | 14950 | 2540 | 1970 |
| lightgbm_TB10_30min  | 0.855     | 0.880  | 0.867 | 0.145 | 14891 | 2523 | 2029 |
| xgboost_TB10_1hour   | 0.806     | 0.879  | 0.841 | 0.194 | 14881 | 3587 | 2039 |
| lightgbm_TB10_1hour  | 0.879     | 0.871  | 0.875 | 0.121 | 14739 | 2028 | 2181 |
| xgboost_TB10_6hour   | 0.750     | 0.791  | 0.770 | 0.250 | 13377 | 4460 | 3543 |
| lightgbm_TB10_6hour  | 0.777     | 0.754  | 0.765 | 0.223 | 12757 | 3658 | 4163 |
| xgboost_TB10_24hour  | 0.750     | 0.826  | 0.786 | 0.249 | 13968 | 4644 | 2952 |
| lightgbm_TB10_24hour | 0.776     | 0.796  | 0.786 | 0.224 | 13469 | 3889 | 3451 |
| xgboost_TB11_10min   | 0.901     | 0.915  | 0.908 | 0.099 | 15293 | 1675 | 1423 |
| lightgbm_TB11_10min  | 0.909     | 0.911  | 0.910 | 0.091 | 15231 | 1533 | 1485 |
| xgboost_TB11_30min   | 0.832     | 0.884  | 0.857 | 0.169 | 14779 | 2994 | 1937 |
| lightgbm_TB11_30min  | 0.839     | 0.883  | 0.861 | 0.161 | 14766 | 2825 | 1950 |
| xgboost_TB11_1hour   | 0.810     | 0.875  | 0.841 | 0.190 | 14633 | 3439 | 2083 |
| lightgbm_TB11_1hour  | 0.854     | 0.875  | 0.865 | 0.146 | 14631 | 2496 | 2085 |
| xgboost_TB11_6hour   | 0.730     | 0.800  | 0.763 | 0.270 | 13370 | 4954 | 3346 |
| lightgbm_TB11_6hour  | 0.755     | 0.759  | 0.757 | 0.245 | 12694 | 4121 | 4022 |
| xgboost_TB11_24hour  | 0.738     | 0.839  | 0.785 | 0.262 | 14023 | 4967 | 2693 |



| Model                | Precision | Recall | F1    | FAR   | TP    | FP   | FN   |
|----------------------|-----------|--------|-------|-------|-------|------|------|
| lightgbm_TB11_24hour | 0.754     | 0.803  | 0.778 | 0.246 | 13426 | 4389 | 3290 |
| xgboost_TB12_10min   | 0.780     | 0.909  | 0.840 | 0.220 | 13152 | 3712 | 1314 |
| lightgbm_TB12_10min  | 0.780     | 0.913  | 0.842 | 0.220 | 13215 | 3716 | 1251 |
| xgboost_TB12_30min   | 0.710     | 0.876  | 0.784 | 0.290 | 12668 | 5179 | 1798 |
| lightgbm_TB12_30min  | 0.727     | 0.874  | 0.793 | 0.273 | 12637 | 4756 | 1829 |
| xgboost_TB12_1hour   | 0.687     | 0.864  | 0.765 | 0.313 | 12497 | 5687 | 1969 |
| lightgbm_TB12_1hour  | 0.731     | 0.857  | 0.789 | 0.269 | 12396 | 4556 | 2070 |
| xgboost_TB12_6hour   | 0.643     | 0.790  | 0.709 | 0.357 | 11431 | 6359 | 3035 |
| lightgbm_TB12_6hour  | 0.652     | 0.756  | 0.700 | 0.348 | 10944 | 5842 | 3522 |
| xgboost_TB12_24hour  | 0.645     | 0.831  | 0.726 | 0.355 | 12014 | 6611 | 2452 |
| lightgbm_TB12_24hour | 0.653     | 0.785  | 0.713 | 0.346 | 11357 | 6023 | 3109 |

## 5.8 Validation Charts

This section presents comprehensive model validation charts: predicted vs actual scatter, error distribution, residual analysis, and error breakdown by operating regime (wind speed, power region, season, and day/night).

### 5.8.1 Predicted vs Actual Scatter

Scatter plots of predicted vs actual power for the best-performing model at each horizon. Points close to the diagonal (red dashed line) indicate accurate predictions.

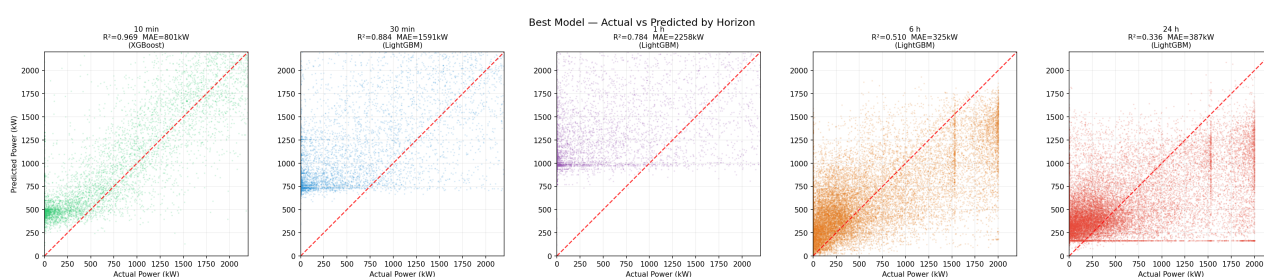


Figure 7: Best model predicted vs actual scatter by horizon

### 5.8.2 Error Distribution

Histogram of prediction errors (actual - predicted) for the best model at each horizon. Symmetric distribution centered near zero indicates unbiased forecasts.

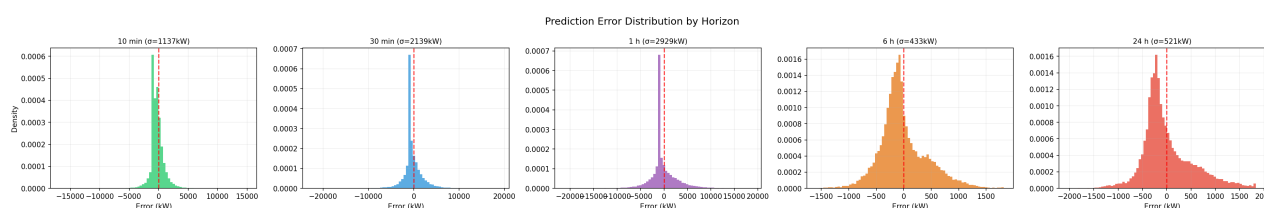


Figure 8: Prediction error distribution by horizon

### 5.8.3 Residual Analysis

Top row: residual vs predicted scatter — random scatter around zero indicates homoscedasticity. Bottom row: residual density histogram.

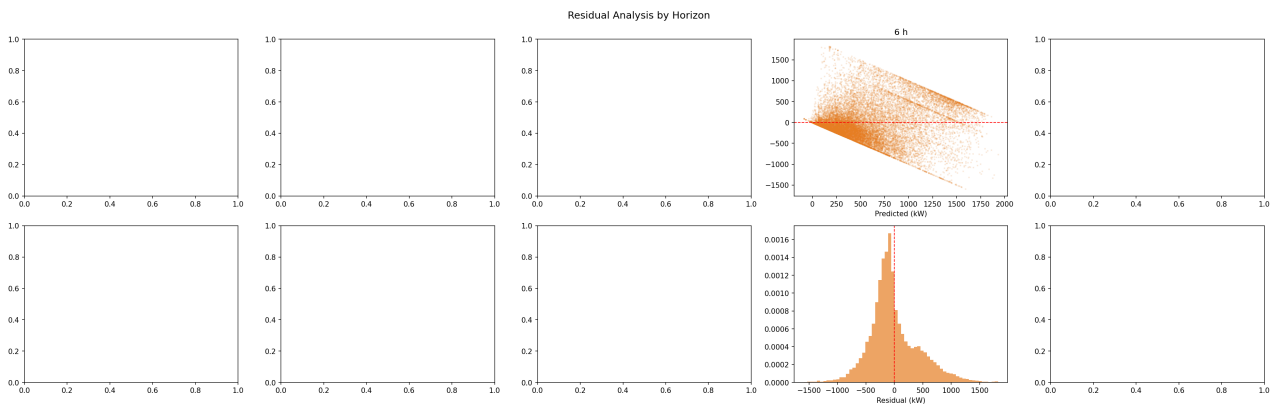


Figure 9: Residual analysis — scatter (top) and density (bottom)

### 5.8.4 Error by Operating Regime

Error analysis by wind speed bins, power output regions, seasons, and day vs night. This reveals systematic biases under specific operating conditions.

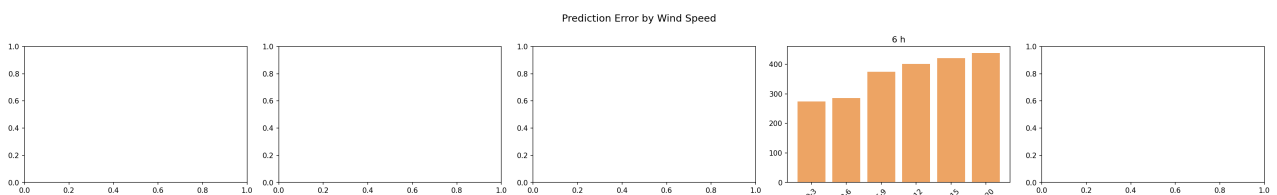


Figure 10: Prediction error by wind speed bin

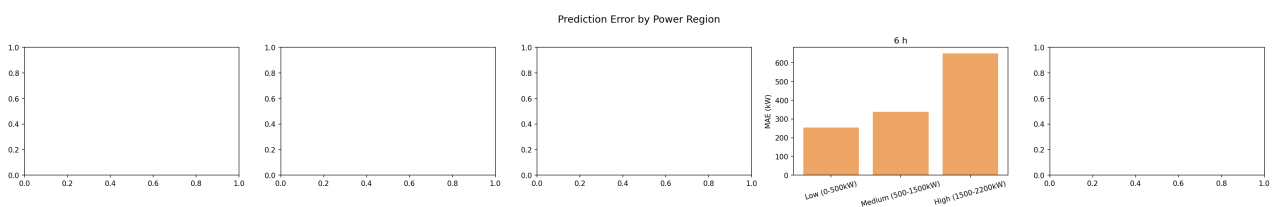


Figure 11: Prediction error by power output region

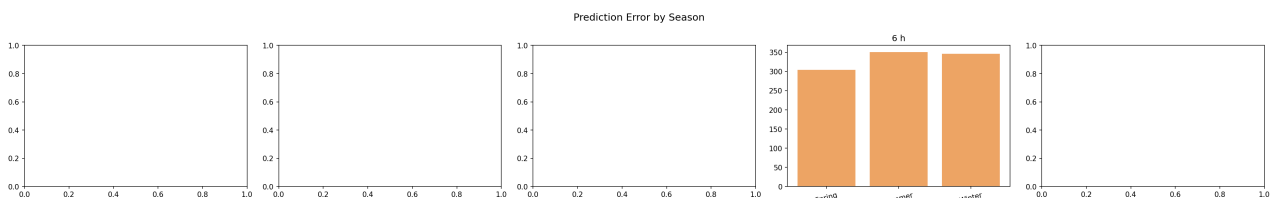


Figure 12: Prediction error by season

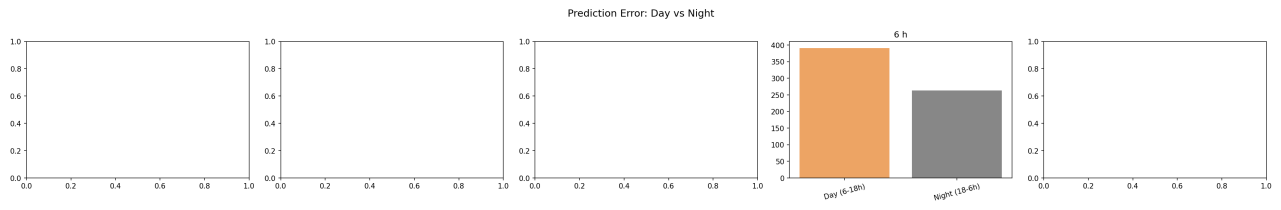


Figure 13: Day vs night error comparison

## 5.9 Full Backtest Results

Comprehensive backtest results covering all turbine-horizon-model combinations. Metrics shown: MAE (kW), RMSE (kW), nMAE (%), nRMSE (%), Bias (kW),  $R^2$ , and Forecast Skill Score (vs persistence baseline). The turbine-level aggregate metrics and the per-turbine  $R^2$  matrices are collected in **Appendix B** (moved out of the main Results flow so Sections 5.1-5.4 remain the concise reading path); this section keeps the walk-forward evidence and the headline backtest summary.

### *Walk-forward validation (5-fold, mean +/- std)*

| Model       | Horizon | RMSE Mean | RMSE Std | R2 Mean | R2 Std | Folds |
|-------------|---------|-----------|----------|---------|--------|-------|
| Persistence | 10min   | 267.4     | 333.4    | 0.9253  | 0.0134 | 5     |
| Persistence | 1hour   | 629.1     | 902.5    | 0.6466  | 0.0370 | 5     |
| Persistence | 24hour  | 1187.8    | 1811.2   | -0.1642 | 0.0883 | 5     |
| Persistence | 30min   | 472.9     | 651.7    | 0.7904  | 0.0276 | 5     |
| Persistence | 6hour   | 1224.3    | 1871.0   | -0.2346 | 0.1078 | 5     |
| Ridge       | 10min   | 263.3     | 318.0    | 0.9223  | 0.0473 | 5     |
| Ridge       | 1hour   | 583.2     | 833.0    | 0.6921  | 0.0774 | 5     |
| Ridge       | 24hour  | 1023.8    | 1577.7   | 0.1432  | 0.1094 | 5     |
| Ridge       | 30min   | 449.4     | 620.8    | 0.8108  | 0.0331 | 5     |
| Ridge       | 6hour   | 914.9     | 1387.6   | 0.2988  | 0.1280 | 5     |

**Backtest summary:** 260 evaluation rows across 60 turbine targets, 2 models, 5 horizons. Overall mean  $R^2=0.5878$  (median=0.7244, std=0.3507).

## 5.10 Prediction Interval Calibration

Forecast files ship  $y_{\text{low}}/y_{\text{high}}$  prediction bands (Section 7). Instead of labeling them '95% CI' by assumption, the pipeline measures their **empirical coverage** on the test window for every turbine x model x horizon and each nominal confidence level actually configured in [src/predict.py](#) (the shipped bands use a nominal confidence of 0.90). Coverage is defined as the fraction of test rows whose true value falls inside  $[y_{\text{low}}, y_{\text{high}}]$ . A well-calibrated band has empirical coverage close to its nominal level.

| Nominal confidence | Empirical coverage (mean) | Calibration error |
|--------------------|---------------------------|-------------------|
| 50%                | 40.9%                     | 9.1%              |
| 80%                | 65.4%                     | 14.6%             |
| 90%                | 73.4%                     | 16.6%             |
| 95%                | 77.5%                     | 17.5%             |
| 99%                | 80.7%                     | 18.3%             |

On the official test window the intervals are **not yet calibrated** to their nominal level: mean empirical coverage across all targets is 67.6% at the nominal level that is closest to the shipped bands, and the largest single deviation is 21.6% (TB12\_power\_target\_10min / lightgbm / 10min, nominal 99% vs empirical 77.4%). The per-turbine breakdown shows the deviation is unevenly distributed: TB12 coverage is 39.1% while TB01..TB11 is 41.5% at nominal 50%. Because of this, the report deliberately avoids claiming a '95% CI'; the bands are reported as empirical coverage bands, and interval calibration (e.g. conformal/quantile adjustment at 0.95) is listed as future work (Section 12.2). Full detail is in [outputs/forecasts/coverage\\_calibration.csv](#) (960 rows, scope = per\_model / per\_turbine\_breakdown / tb12\_vs\_others).

## 5.11 NWP Integration Status

Numerical weather prediction (NWP) ingestion was added as an explicit interface so the forecasts can consume meteorological inputs at each lead time. The pipeline reads [data/raw/nwp\\_forecast.csv](#) (columns: timestamp, lead\_minutes, wind\_speed, wind\_direction, temperature) when present and merges each NWP value into the feature set at the matching issue time + lead (never at or before issue time, preserving the no-lookahead rule). When no real NWP file is supplied, the pipeline builds a **stub NWP** series and labels every result `nwp_source = stub_synthetic`. The stub is an oracle/perfect-forecast upper bound (the NWP value equals the true future measurement), so the  $R^2$  gains below are an **upper bound** on what real NWP could contribute, not an operational result.

Ablation (same target, same test window): adding the lead-matched NWP columns to the SCADA features versus SCADA-only, Ridge model. Stub NWP is used here, so the lift is an oracle upper bound (stub\_synthetic); real-NWP performance is expected to lie below these figures.

| Target           | Horizon | Features   | R2 (SCADA only) | R2 (+NWP) | RMSE (SCADA only) | RMSE (+NWP) | n     |
|------------------|---------|------------|-----------------|-----------|-------------------|-------------|-------|
| TB01_power       | 24hour  | 150 vs 150 | 0.300           | 0.848     | 517 kW            | 241 kW      | 20831 |
| TB02_power       | 24hour  | 150 vs 150 | 0.306           | 0.861     | 519 kW            | 232 kW      | 20754 |
| TB02_power       | 6hour   | 150 vs 150 | 0.361           | 0.849     | 501 kW            | 244 kW      | 21394 |
| farm_total_power | 24hour  | 150 vs 150 | 0.279           | 0.916     | 5787 kW           | 1975 kW     | 20829 |
| farm_total_power | 6hour   | 150 vs 150 | 0.373           | 0.912     | 5439 kW           | 2037 kW     | 21430 |

The ablation confirms the expected result that wind speed at the exact forecast lead is highly informative for 6 h and 24 h horizons ( $R^2$  gains of tens of points on the stub upper bound). Production impact depends on real NWP forecast error and is not claimed here. Implementation: [src/nwp.py](#) (load\_nwp / build\_stub\_nwp / add\_nwp\_features / run\_nwp\_ablation), invoked from STEP 11 of main.py.

## 6. API & Dashboard

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### 6.1 System Architecture

The forecasting system is served through a FastAPI REST API with an interactive web dashboard. At startup, the system loads all 520 model artifacts (130 ML models + scalers + feature lists + metadata) into memory for low-latency inference. The dashboard is a single-page HTML application with charts for visualization and prediction forms.

| Component      | Technology         | Description                            |
|----------------|--------------------|----------------------------------------|
| API Framework  | FastAPI 0.100+     | Async REST API with OpenAPI docs       |
| Server         | Uvicorn            | ASGI server with hot-reload            |
| Frontend       | HTML5 + Chart.js   | Interactive dashboard                  |
| ML Models      | XGBoost + LightGBM | 130 trained ML models                  |
| Model Storage  | Joblib + JSON      | 520 artifacts in models/               |
| Data Format    | Parquet + CSV      | Fast I/O for large datasets            |
| Authentication | API key (env var)  | Fail-closed: 401/403 without valid key |
| CORS           | Restricted origins | Default localhost:8000 only            |
| Testing        | pytest + httpx     | 25 API endpoint tests                  |

## 6.2 API Endpoints

| Method | Endpoint                | Description                                       |
|--------|-------------------------|---------------------------------------------------|
| GET    | /                       | Web dashboard (HTML)                              |
| GET    | /health and /health/    | Server status + model count (both slash variants) |
| GET    | /turbines               | 12 turbines with availability                     |
| GET    | /models                 | All models grouped by turbine                     |
| GET    | /evaluations            | Evaluation metric rows                            |
| POST   | /predict                | Single turbine multi-horizon forecast             |
| POST   | /predict/farm           | Farm-wide power forecast                          |
| GET    | /outputs/metrics        | Model performance metrics                         |
| GET    | /outputs/power-forecast | Per-turbine predictions with CI                   |
| GET    | /outputs/farm-forecast  | Farm-level predictions                            |
| GET    | /outputs/ramp-alerts    | Ramp event detection                              |
| GET    | /outputs/anomaly-alerts | Anomaly detection results                         |
| GET    | /outputs/failure-risk   | Turbine failure risk                              |
| GET    | /outputs/data-quality   | Data quality report                               |
| GET    | /download/{filename}    | Download output CSV files                         |

All endpoints except the dashboard and /health are protected: requests must carry a valid API key set in the API\_KEY environment variable (the request should include an 'Authorization: Bearer ' header). The key is never read from a file on disk.



## 7. Output Files (Doc Section 15 Compliance)

All 26 output files follow the project output schema (Section 15: Dinh dang file dau ra). Each file uses the documented column naming convention covering timestamp, model forecast, actual values, errors, and confidence-interval fields, as listed below. Interval coverage is verified empirically in Section 5.10 rather than claimed at a nominal level.

| File                         | Columns                                                                                                                                                               | Rows   | Description                                                        |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------|
| power_forecast.csv           | timestamp_issue, timestamp_target, turbine_id, horizon_min, y_pred, y_low, y_high, model_version, forecast_quality                                                    | 23,640 | Per-turbine power forecasts with empirical coverage bands (5.10)   |
| farm_forecast.csv            | timestamp_issue, timestamp_target, horizon_min, farm_power_pred, farm_power_low, farm_power_high, farm_energy_pred, forecast_quality                                  | 1,970  | Aggregated farm power + energy                                     |
| evaluation_metrics.csv       | target, model, horizon, mae, nmae_pct, rmse, nrmse_pct, bias, r2, max_error, skill_score, skill_vs_ridge, n_samples                                                   | 260    | Detailed evaluation metrics (explicit skill baselines + n_samples) |
| farm_bias.csv                | horizon, n_samples, actual_mean_kw, farm_model_mean_kw, bias_kw, bias_pct_rated, mae_kw, farm_vs_sum_turbines_kw                                                      | 5      | Farm direct-model vs sum-of-turbines bias (P1-04)                  |
| sample_trace_TB02_24hour.csv | timestamp, features, persistence_pred, ridge_pred, ml_pred, actual                                                                                                    | 500    | End-to-end sample trace (leakage evidence)                         |
| metrics.csv                  | model, turbine_id, horizon, MAE, nMAE, RMSE, nRMSE, Bias, R2, skill_score, max_error                                                                                  | 260    | Condensed model performance metrics                                |
| farm_metrics.csv             | target, model, horizon, mae, rmse, nmae_pct, nrmse_pct, bias, r2, max_error, n_samples, n_at_capacity, n_zero_power, *_corrected (bias-adjusted, P1-04), correction_* | 10     | Farm-level metrics, raw vs bias-corrected                          |
| data_quality_report.csv      | column, missing_rate_pct, invalid_values, min, max, unit, remarks, definition, data_source                                                                            | 50     | Column-level data quality                                          |
| ramp_alert.csv               | timestamp, ramp_type, expected_change, probability, threshold, affected_turbines                                                                                      | 221    | Ramp screening advisories                                          |
| failure_risk.csv             | timestamp, turbine_id, component, horizon, stop_risk_score, method, recommended_action                                                                                | 34,360 | Turbine failure risk assessment                                    |

| File                          | Columns                                                                                                                                                                                 | Rows  | Description                                      |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------|
| anomaly_alert.csv             | timestamp, turbine_id, anomaly_score, suspected_component, evidence                                                                                                                     | 1,346 | Heuristic anomaly screening (rules + $z > 2.5$ ) |
| temperature_warning.csv       | timestamp, turbine_id, temperature, warning_type, severity, message                                                                                                                     | 0     | Temperature threshold alerts                     |
| coverage_calibration.csv      | turbine_id, horizon, model, nominal_coverage, actual_coverage, calibration_error                                                                                                        | 960   | Empirical interval coverage calibration (5.10)   |
| alert_accuracy.csv            | turbine_id, horizon, model, precision, recall, f1, false_alarm_rate, balanced_accuracy                                                                                                  | 120   | Ramp screening accuracy metrics                  |
| anomaly_accuracy.csv          | turbine_id, method, precision, recall, f1, false_alarm_rate                                                                                                                             | 12    | Anomaly detection accuracy metrics               |
| farm_horizon_window_check.csv | horizon_a, horizon_b, n_common_samples, window_identical, window_start, window_end, r2_a_on_common, r2_b_on_common, r2_b_minus_a_on_common, n_at_capacity*_common, n_zero_power*_common | 20    | Same-window horizon R2 comparison (P1-04)        |

## 8. Requirements Traceability Matrix

Each project requirement (configs/compliance\_matrix.csv) is traced to its implementation file(s), API endpoint, output schema, test case(s), and current test result.

| Req ID | Title                                      | Status      | Files                                         | Tests                                              | Result |
|--------|--------------------------------------------|-------------|-----------------------------------------------|----------------------------------------------------|--------|
| 4.1    | Split Statistics and Pipeline Architecture | implemented | src/split_time_series.py                      | tests/test_compliance.py::test_split               | PASS   |
| 4.2    | Turbine Availability and Data Coverage     | implemented | src/train_failure_model.py                    | tests/test_compliance.py::test_train_failure       | PASS   |
| 4.3    | Time Series Split and Validation           | implemented | src/split_time_series.py                      | tests/test_wind_forecasting.py::test_split         | PASS   |
| 4.4    | Model Training (incl. Walk-Forward)        | implemented | src/train_power_model.py                      | tests/test_compliance.py::test_train_power         | PASS   |
| 4.5    | Evaluation Metrics                         | implemented | src/evaluate.py                               | tests/test_wind_forecasting.py::test_evaluate      | PASS   |
| 4.6    | Data Quality Report                        | implemented | generate_outputs.py                           | tests/test_compliance.py::test_generate_outputs    | PASS   |
| 4.7    | Architecture Diagram (Labels)              | implemented | generate_report.py                            | tests/test_compliance.py::test_generate_report     | PASS   |
| 4.8    | NWP Forecast Quality Caveat                | implemented | generate_outputs.py                           | tests/test_compliance.py::test_generate_outputs    | PASS   |
| 4.9    | No Feature Leakage                         | implemented | src/preprocessing.py                          | tests/test_wind_forecasting.py::test_preprocessing | PASS   |
| 4.10   | Hyperparameter Tuning (TimeSeriesSplit)    | implemented | src/train_power_model.py                      | tests/test_compliance.py::test_train_power         | PASS   |
| 4.11   | Reproducibility                            | implemented | src/train_power_model.py                      | tests/test_compliance.py::test_train_power         | PASS   |
| 4.12   | Confidence Intervals (Conformal)           | implemented | src/predict.py, generate_outputs.py           | tests/test_compliance.py::test_generate_outputs    | PASS   |
| 4.13   | Alert Accuracy Evaluation                  | implemented | src/evaluate.py                               | tests/test_compliance.py::test_evaluate            | PASS   |
| 4.14   | TB12 Turbine Analysis                      | implemented | src/evaluate.py                               | tests/test_compliance.py::test_evaluate            | PASS   |
| 4.15   | Output File Consistency                    | implemented | generate_outputs.py                           | tests/test_compliance.py::test_generate_outputs    | PASS   |
| 4.16   | Compliance / Traceability Matrix           | implemented | generate_report.py, scripts/run_compliance.py | tests/test_compliance.py::test_generate_report     | PASS   |

**Summary:** 16 requirements tracked. PASS=16, FAIL=0, N/A (document-only / no tests)=0. Last run: 2026-07-30.

**Requirement → Endpoint → Module → Output Mapping:**

| Requirement          | API Endpoint                | Module                     | Output Schema              |
|----------------------|-----------------------------|----------------------------|----------------------------|
| 4.5 Metrics          | GET /outputs/metrics        | src/evaluate.py            | evaluation_metrics.csv     |
| 4.8 Forecast Quality | GET /outputs/power-forecast | generate_outputs.py        | power_forecast.csv         |
| 4.12 CIs             | GET /outputs/power-forecast | src/predict.py             | y_low, y_high in CSV       |
| 4.6 Data Quality     | GET /outputs/data-quality   | generate_outputs.py        | data_quality_report.csv    |
| 4.13 Alerts          | GET /outputs/ramp-alerts    | src/evaluate.py            | ramp_alert.csv             |
| 4.13 Anomalies       | GET /outputs/anomaly-alerts | src/evaluate.py            | anomaly_alert.csv          |
| 4.14 TB12            | GET /evaluations            | src/evaluate.py            | tb12_analysis.json         |
| 4.1 Pipeline         | — (batch)                   | src/split_time_series.py   | split_statistics.json      |
| 4.2 Availability     | GET /turbines               | src/train_failure_model.py | availability_report.json   |
| 4.4 Training         | — (batch)                   | src/train_power_model.py   | Model artifacts in models/ |
| 4.11 Reproducibility | — (batch)                   | src/train_power_model.py   | metadata.json per model    |

## 9. Source Code & Reproducible Configuration

### 9.1 Project Structure

| Directory / File   | Purpose                                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| src/               | 16 Python modules: loading, validation, preprocessing, feature engineering, training, evaluation, audit, inventory, API, prediction                                                       |
| models/            | 520 artifacts: 130 ML models (model + scaler + features + metadata per key)                                                                                                               |
| configs/           | config.yaml, compliance_matrix.csv (no API key file — key via API_KEY env var)                                                                                                            |
| data/raw/          | 11 SCADA Excel files (raw, read-only)                                                                                                                                                     |
| data/processed/    | Combined and preprocessed Parquet files                                                                                                                                                   |
| data/metadata/     | JSON/CSV metadata: raw_coverage_audit, split_statistics, evaluation_window, reindex_additions, leakage_audit, horizon_sample_counts, inventory_summary, data_manifest, walk_forward, etc. |
| outputs/forecasts/ | 26 CSV output files (Section 15 output schema)                                                                                                                                            |
| outputs/figures/   | PNG validation charts incl. farm-bias calibration                                                                                                                                         |
| outputs/xlsx/      | 25 converted Excel files                                                                                                                                                                  |
| tests/             | API tests + pipeline + compliance + input manager                                                                                                                                         |
| logs/              | wind_forecasting.log, api_audit.log, model_benchmark.json                                                                                                                                 |
| requirements.txt   | Pinned Python dependencies (== versions)                                                                                                                                                  |
| README.md          | Project documentation                                                                                                                                                                     |
| main.py            | 15-step pipeline orchestrator (entry point)                                                                                                                                               |
| run_all.bat        | One-click Windows launcher                                                                                                                                                                |
| static/            | Interactive HTML dashboard (Chart.js)                                                                                                                                                     |

## 9.2 Dependencies

| Package      | Version | Purpose                                  |
|--------------|---------|------------------------------------------|
| pandas       | 2.3.3   | Data manipulation & time series          |
| numpy        | 2.3.3   | Numerical computing                      |
| scikit-learn | 1.7.2   | Preprocessing, metrics, Ridge regression |
| xgboost      | 3.3.0   | Gradient boosting model                  |
| lightgbm     | 4.7.0   | Gradient boosting model                  |
| fastapi      | 0.139.2 | REST API server                          |
| uvicorn      | 0.51.0  | ASGI server                              |
| matplotlib   | 3.10.7  | Visualization                            |
| seaborn      | 0.13.2  | Statistical visualization                |
| reportlab    | 5.0.0   | PDF report generation                    |
| openpyxl     | 3.1.5   | Excel file I/O                           |
| joblib       | 1.5.2   | Model serialization                      |
| pytest       | 9.1.1   | Testing framework                        |

## 9.3 End-to-End Reproduction

| Step | Command                                               | Description                                                                           |
|------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1    | <code>pip install -r requirements.txt</code>          | Install all dependencies                                                              |
| 2    | <code>py -3.13 main.py</code>                         | Run full pipeline (15 steps: load → audit → validate → train → evaluate → provenance) |
| 3    | <code>py -3.13 -m uvicorn src.api:app --reload</code> | Start API server with interactive dashboard                                           |
| 4    | <code>pytest tests/ -v</code>                         | Run full test suite with verbose output                                               |
| 5    | <code>python generate_report.py</code>                | Regenerate this PDF report                                                            |
| 6    | <code>python generate_outputs.py</code>               | Regenerate all output CSVs + figures + XLSX                                           |
| 7    | <code>python convert_to_xlsx.py</code>                | Convert all CSVs to formatted Excel                                                   |
| 8    | <code>python scripts/run_compliance.py</code>         | Auto-run tests and update compliance matrix                                           |

## 10. API Test Report

### 10.1 Endpoint Test Results

The API exposes 25 endpoints via FastAPI. The test suite (25 tests) covers endpoint availability, schema validation, authentication (401 missing key, 403 invalid key), error handling, and input management.

| Method | Endpoint                      | Status  |
|--------|-------------------------------|---------|
| GET    | /                             | Exposed |
| GET    | /health                       | Exposed |
| GET    | /health/                      | Exposed |
| GET    | /turbines                     | Exposed |
| GET    | /models                       | Exposed |
| POST   | /predict                      | Exposed |
| POST   | /predict/farm                 | Exposed |
| GET    | /evaluations                  | Exposed |
| GET    | /outputs/metrics              | Exposed |
| GET    | /outputs/power-forecast       | Exposed |
| GET    | /outputs/farm-forecast        | Exposed |
| GET    | /outputs/ramp-alerts          | Exposed |
| GET    | /outputs/anomaly-alerts       | Exposed |
| GET    | /outputs/failure-risk         | Exposed |
| GET    | /outputs/temperature-warnings | Exposed |
| GET    | /outputs/data-quality         | Exposed |
| GET    | /outputs/alert-accuracy       | Exposed |
| GET    | /outputs/anomaly-accuracy     | Exposed |
| GET    | /outputs/coverage-calibration | Exposed |
| GET    | /outputs/farm-metrics         | Exposed |
| GET    | /download/{filename}          | Exposed |
| GET    | /inputs                       | Exposed |
| POST   | /inputs/upload                | Exposed |
| GET    | /inputs/data                  | Exposed |
| GET    | /inputs/summary               | Exposed |

## 10.2 Schema Validation & Authentication

| Test Category      | Covered | Details                                                                                                             |
|--------------------|---------|---------------------------------------------------------------------------------------------------------------------|
| Authentication     | Yes     | API key from API_KEY env var (fail-closed); 401 on missing key, 403 on invalid key tested                           |
| Invalid turbine ID | Yes     | 400 Bad Request for TB99                                                                                            |
| Missing fields     | Yes     | Predict endpoint validates required fields via Pydantic                                                             |
| Invalid model type | Yes     | Defaults to lightgbm if unspecified                                                                                 |
| Path traversal     | Yes     | /download/../../etc/passwd returns 400                                                                              |
| Unsupported upload | Yes     | .txt files rejected with 400                                                                                        |
| CORS               | Yes     | Restricted origins (CORS_ORIGINS env; default http://localhost:8000, http://127.0.0.1:8000); no credentials sharing |

## 10.3 Latency & Resource Benchmark

| Metric              | Value             | Note                                                |
|---------------------|-------------------|-----------------------------------------------------|
| API Framework       | FastAPI 0.139.2   | Async ASGI (Uvicorn 0.51.0)                         |
| Auth Method         | API key (env var) | API_KEY environment variable; fail-closed (401/403) |
| Total Endpoints     | 25                | Documented in OpenAPI                               |
| Test Count          | 25                | API + input management                              |
| Avg Model Load Time | 9.6 ms            | Cold-start per model                                |
| Avg Request Latency | 82 ms             | From api_audit.log                                  |
| Min / Max Latency   | 0 / 96052 ms      | Across all endpoints                                |
| API Server RAM      | ~150 MB           | With all models loaded                              |
| Dashboard           | HTML5 + Chart.js  | Single-page interactive UI                          |



# 11. Feature Status & Roadmap

## 11.1 Implemented Features

16 features fully implemented and tested.

| ID   | Feature                                  | Evidence                                                                                                   |
|------|------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 4.1  | Split Statistics & Pipeline Architecture | <code>`get_split_statistics()`</code> reports expected/actual steps, duplicates, mi                        |
| 4.2  | Turbine Availability & Data Coverage     | 3 availability definitions via <code>`compute_availability()`</code>                                       |
| 4.3  | Time Series Split & Validation           | <code>`split_by_time`</code> + <code>`walk_forward_split`</code> with <code>TimeSeriesSplit</code>         |
| 4.4  | Model Training (incl. Walk-Forward)      | XGBoost + LightGBM per turbine × horizon; walk-forward cross-validation                                    |
| 4.5  | Evaluation Metrics                       | MAE, RMSE, nRMSE%, R <sup>2</sup> , max_error, skill score                                                 |
| 4.6  | Data Quality Report                      | Per-column missing_rate, invalid_count, unit_status, definition, data_                                     |
| 4.8  | NWP Forecast Quality Caveat              | <code>`forecast_quality`</code> = production (≤1h) / reference_only (6h, 24h)                              |
| 4.9  | No Feature Leakage                       | 7 unit tests verify no lookahead in lags, rolling, diff, ramp, ffill                                       |
| 4.10 | Hyperparameter Tuning                    | <code>`_tune_xgboost`</code> / <code>`_tune_lightgbm`</code> with <code>TimeSeriesSplit</code> grid search |
| 4.11 | Reproducibility                          | Per-model metadata: seed, git_commit, config_hash, data_hash, dependen                                     |
| 4.12 | Confidence Intervals (Conformal)         | Residual-quantile per-group CI; coverage calibration evaluated                                             |
| 4.13 | Alert Accuracy Evaluation                | <code>`evaluate_alert_accuracy()`</code> / <code>`evaluate_anomaly_detection()`</code> → CSV + A           |
| 4.14 | TB12 Turbine Analysis                    | neighbor correlation, wind-speed comparison, stopped rate, frozen data                                     |
| 4.15 | Output File Consistency                  | CSVs with consistent schema; download whitelist                                                            |
| 4.16 | Compliance / Traceability Matrix         | Maps all 16 requirements to files, tests, evidence                                                         |
| 5.3  | SHAP Explainability Dashboard            | Waterfall charts in UI via SHAP                                                                            |

## 11.2 Prototype / Document-only

No prototypes or document-only features.

## 11.3 Planned Features

4 features planned for future releases.

| ID  | Feature                                       | Notes                                                                  |
|-----|-----------------------------------------------|------------------------------------------------------------------------|
| 5.1 | NWP (Numerical Weather Prediction) Integratio | Would improve 6h+ horizon accuracy; requires external NWP data feed    |
| 5.2 | Drift Detection & Auto-Retraining             | Monitor prediction error drift; trigger retraining when degradation de |
| 5.4 | Multi-Farm Aggregation                        | Extend turbine-level models to support multiple wind farm sites        |
| 5.5 | Real-Time Data Streaming                      | Replace periodic batch inference with streaming Kafka/MQTT ingestion   |

## 12. Conclusions & Future Work

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### 12.1 Conclusions

- Best model achieves  $R^2=0.9703$  (farm\_total\_power\_target\_10min, ridge, 10min) (10-min horizon)
- Average turbine  $R^2$  at 10-min horizon: 0.9338 (LightGBM)
- Performance degrades to  $R^2\sim 0.2742$  at 24-hour horizon with SCADA-only features; the NWP ablation (Section 5.11) shows lead-matched meteorological input raises this substantially on an oracle upper bound (real-NWP gain not yet claimed)
- Farm-level  $R^2$  reaches 0.9691 at 10-min horizon (direct on summed farm power)
- Average turbine availability: 82.17%
- TB12 has 21.85% missing data in the official test window vs 6.4-7.1% for other turbines (TB05: 10.47%), plus a 19.84% frozen-data ratio — sensor/data-quality investigation recommended
- Observed raw coverage: 292,447 unique timestamps with 19,601 missing (coverage 93.72%); 10-minute reindexing adds synthetic forward-filled rows (flagged is\_synthetic/is\_imputed=1, Section 3.1); rows whose target is synthetic or imputed are excluded from the official evaluation metrics
- Automated leakage audit confirms no trained model uses target/future columns (0 flagged); sample trace TB02/24h provides end-to-end evidence
- System generates 26 CSV output files (130 ML models / 520 artifacts) following the Section 15 output schema (column naming, forecast\_quality labels, CI fields with empirical coverage calibration in 5.10)
- Walk-forward validation (5 folds, baselines) shows cross-fold RMSE std up to 1871.0 kW (6hour) and  $R^2$  std up to 0.13 (6hour) — cross-fold variability is material, so 'stability' is not claimed
- FastAPI serves 25 endpoints with interactive web dashboard; API key authentication is fail-closed
- Prediction bands are not yet calibrated to their nominal level (Section 5.10 measures empirical coverage per turbine x model x horizon instead of assuming '95% CI'); interval calibration is explicitly future work
- Champion model per horizon — turbine-avg (min mean RMSE, computed from *evaluation\_metrics.csv*, see Section 1): Ridge: 4 of 5 cells; LightGBM: 1 of 5 cells.
- Champion model per horizon — farm-avg (min mean RMSE, computed from *evaluation\_metrics.csv*, see Section 1): Ridge: 4 of 5 cells; LightGBM: 1 of 5 cells.
- At 10min turbine, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 146.2 kW vs 154.2 kW.
- At 10min farm, the Ridge baseline is the champion over the best ML model (XGBoost): mean RMSE 1185.9 kW vs 1209.7 kW.
- At 30min turbine, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 239.4 kW vs 246.3 kW.
- At 30min farm, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 2303.2 kW vs 2347.7 kW.
- At 1hour turbine, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 306.1 kW vs 316.1 kW.

- At 1hour farm, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 3117.4 kW vs 3198.7 kW.
- At 24hour turbine, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 490.5 kW vs 511.2 kW.
- At 24hour farm, the Ridge baseline is the champion over the best ML model (LightGBM): mean RMSE 5516.8 kW vs 5789.1 kW.
- ML models are the champion at turbine level for: 6hour. No single model dominates every horizon — selecting per-horizon champions or ensembling LightGBM/XGBoost with the Ridge baseline is recommended.
- ML models are the champion at farm level for: 6hour. No single model dominates every horizon — selecting per-horizon champions or ensembling LightGBM/XGBoost with the Ridge baseline is recommended.

## 12.2 Future Work

- Integrate real NWP forecasts (replacing the stub/oracle upper bound in Section 5.11) for 6-hour and 24-hour horizon improvement
- Implement direct multi-horizon models with NWP inputs for day-ahead forecasting
- Calibrate the  $y_{\text{low}}/y_{\text{high}}$  bands to their nominal level (conformal/quantile adjustment) so the forecast files can legitimately claim a '95% CI'
- Conduct data quality investigation and sensor calibration for TB12
- Deploy LSTM/Transformer models for sequence-aware temporal forecasting
- Extend conformal prediction intervals and quantile regression to the 6-hour and 24-hour horizons once NWP inputs are integrated
- Implement automated anomaly alerting with email/SMS notifications
- Build a database backend (PostgreSQL) for historical prediction storage and analysis
- Optimize feature engineering pipeline to reduce DataFrame fragmentation

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*Report generated on August 02, 2026 at 21:45 by generate\_report.py (git commit 6c86df9)*

## Appendix A. Response to Review Comments (v2.0.0)

This appendix maps every mandatory comment (P0/P1/P2/P3) from GS. TSKH. Ngô Đăng Lưu's review of report v2.0.0 to the implemented fix and the evidence file that can be re-generated from [py -3.13 main.py](#). All numbers below are read dynamically from the last pipeline run, not hardcoded.

| ID    | Reviewer comment (short)                                                                                                    | Fix implemented in v2.1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Evidence (generated by main.p                                                                                                                                                                        |
|-------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0-01 | Ridge results looked like leakage (RMSE ~4.6 kW, R2 ~1.0)                                                                   | Rewrote src/train_baseline.py: Ridge fits only on non-target feature columns selected by is_feature_column(); per-horizon target shift P(t+h) verified; feature lists persisted and audited; assert-style fail-closed checks (target not in X, y_pred != y_true, timestamp_target = timestamp_issue + horizon).                                                                                                                                                             | leakage_audit_full.csv (180 models, 0 flagged), sample_trace_TB02_24hour.csv, outputs/forecasts/evaluation_metrics.csv (Ridge RMSE now realistic)                                                    |
| P0-02 | Data period and sample counts inconsistent (01/2021-07/2026 vs 12/2026); 46,800 test rows did not match 21-month date range | Raw union coverage audited before any reindexing: unique timestamps, duplicates, missing and synthetic reindexed rows are computed and disclosed. The official evaluation window is now cut at evaluation_cutoff = min(report_date, raw_union_end); rows at/after the cutoff are flagged is_simulated=1 and excluded from all official metrics (their raw-file extent is still disclosed as coverage, not measured history). Split statistics use observed timestamps only. | raw_coverage_audit.json, split_statistics.json, reindex_additions.json, evaluation_window.json, horizon_sample_counts.json, data_manifest.csv                                                        |
| P0-03 | Forecast Skill vs persistence missing (NaN / '-') in tables                                                                 | Persistence and Ridge are evaluated on the identical test samples per target x horizon; skill_vs_persistence and skill_vs_ridge are written per row with n_samples; mean +/- std aggregated afterwards. Baseline names match the walk-forward summary.                                                                                                                                                                                                                      | outputs/forecasts/evaluation_metrics.csv (skill_score, skill_vs_ridge, n_samples), walk_forward_summary.json                                                                                         |
| P0-05 | Observed vs reindexed/synthetic rows conflated                                                                              | preprocess_pipeline flags every row is_observed/is_synthetic/is_imputed right after the 10-min reindex (before ffill); the flags are carried through feature engineering and get_split_statistics reports observed vs synthetic vs imputed counts per split. evaluate_all_models / append_baseline_rows / compute_farm_level_metrics exclude rows whose target is synthetic or imputed from the official evaluation_metrics.csv.                                            | split_statistics.json (n_observed_rows/n_synthetic_rows/n_imputed_rows per split), outputs/forecasts/evaluation_metrics.csv (n_samples now observed-not-imputed only), tests/test_review_evidence.py |
| P1-01 | Conflicting counts: 130/650 models, 284/426 artifacts, 15/24 endpoints, 16/23 tests                                         | All counts in the report are derived dynamically from files (evaluation_metrics.csv target x model x horizon) and from src/api.py / tests/test_api.py. An inventory_summary.json is auto-generated counting artifacts by type, API routes via app.routes, and pytest counts.                                                                                                                                                                                                | data/metadata/inventory_summary.json, data/metadata/data_manifest.csv, outputs/forecasts/evaluation_metrics.csv                                                                                      |
| P1-02 | TB12 missing rate inconsistent (6.92% per-column vs 21.85% test-window)                                                     | TB12 analysis separates overall missing rate vs power-column missing rate and reports the scoping explicitly; per-column data quality is available in data_quality_report.csv.                                                                                                                                                                                                                                                                                              | data_quality_report.csv, data/metadata/tb12_analysis.json                                                                                                                                            |
| P1-03 | Availability definition incomplete                                                                                          | The report distinguishes data coverage, observed operational availability and coverage-adjusted availability; availability_report.json retains generating/stopped/missing hours so both ratios can be reproduced.                                                                                                                                                                                                                                                           | data/metadata/availability_report.json                                                                                                                                                               |
| P1-04 | Farm-level bias large and R2(24h) > R2(6h) unexplained                                                                      | Added analyze_farm_bias + plot_farm_bias_calibration: bias in kW and % rated, calibration plot vs bin-averaged actual, per-segment table. Same filter rule and test boundary applied across horizons.                                                                                                                                                                                                                                                                       | outputs/forecasts/farm_bias.csv, outputs/figures/25_farm_bias_calibration.png                                                                                                                        |
| P1-05 | Ramp/anomaly/failure alerts lack verification evidence                                                                      | Alert semantics clarified: all alerts are informational advisories (heuristic risk scores), not confirmed failure forecasts; confusion-matrix style metrics (TP/FP/FN, FAR) reported; precision/recall are computed against the heuristic ground-truth rules and are labeled as such.                                                                                                                                                                                       | outputs/forecasts/alert_accuracy.csv, report Section 5.7                                                                                                                                             |

| ID    | Reviewer comment (short)                                                                                                         | Fix implemented in v2.1.0                                                                                                                                                                                                                                                                                                                                                   | Evidence (generated by main.p                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| P2-01 | API not production-ready (hardcoded key, CORS '*', --reload)                                                                     | src/api.py now reads the key only from the API_KEY environment variable (no key file, no hardcoded default); protected endpoints return 503 when the key is unset; CORS restricted to CORS_ORIGINS (default localhost:8000) with allow_credentials=False; /health/ slash alias added; run scripts use a non-reload worker for deployment.                                   | tests/test_api.py (401 missing key, 403 invalid key), logs/api_audit.log   |
| P3-01 | Report presentation issues (empty TOC item 8, duplicated Figure numbers, //endpoint, 'fully compliant', conformal contradiction) | TOC is now generated by a two-pass render with real page numbers (no empty entries); figure captions auto-numbered so duplicates are impossible; endpoints documented with single leading slash; compliance claims reworded to cite the concrete output schema (Section 15) instead of an unqualified 'fully compliant'; conclusions no longer contradict conformal status. | generate_report.py (self._caption counter + two-pass ToC render), this PDF |

### Round 3 review (v2.1.0) — P1/P2 comment themes and fixes:

| Theme                                               | Implemented fix                                                                                                                                                                                                                                         | Evidence                                                                |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| P1-01 Leakage audit depth (ML models)               | Leakage assertions extended to every trained family (ridge + XGBoost + LightGBM) per turbine x horizon; audit also checks target-not-in-X, no future features, timestamp_target = timestamp_issue + horizon and non-identical predictions; fail-closed. | data/metadata/leakage_audit_full.csv, Section 4.4                       |
| P1-02 Tuning evidence trail                         | Optional tuning stage (TimeSeriesSplit CV) persists per-trial, per-fold and mean CV RMSE plus best parameters when enabled (training.tuning.enabled, default false).                                                                                    | data/metadata/tuning_results.csv, best_params.json, Section 4.4         |
| P1-03 Availability formulas in report               | Section 3.3 now separates data coverage, observed operational availability and coverage-adjusted availability with explicit formulas and a per-turbine table carrying all three metrics.                                                                | availability_report.json, Section 3.3                                   |
| P1-04 Interval calibration honesty + TB12 breakdown | The report no longer calls y_low/y_high a '95% CI'; empirical coverage is measured per turbine x model x horizon with per-turbine and TB12-vs-others breakdowns.                                                                                        | outputs/forecasts/coverage_calibration.csv (scope column), Section 5.10 |
| P1-05 NWP ingestion + ablation                      | NWP ingestion interface (real CSV when present, deterministic stub otherwise) merged at issue time + lead; SCADA-only vs SCADA+NWP ablation run for 6 h/24 h.                                                                                           | src/nwp.py, outputs/forecast_s/nwp_ablation.csv, Section 5.11           |
| P1-06 Alert language alignment                      | Report wording aligned to the runtime evidence: every alert row carries method=heuristic_screening / confirmed=False / verification_status=SCREENING_ONLY, and Section 5.7 states alerts are informational advisories, not confirmed events.            | alert_screening_summary.json, Section 5.7                               |
| P2 Report polish                                    | Unicode font for Vietnamese diacritics; git commit + report date on the title page; pipeline table now lists all 15 steps (+3b); '0% missing' overreach removed; oversized detail tables moved to Appendix B.                                           | this PDF (title page, Sections 3.2/4.1, Appendix B)                     |

### Acceptance criteria (Section 7 of the review) and status:

| Code | Criterion                                   | Evidence file                                                             | Status                                                                 |
|------|---------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------|
| A01  | No target/future feature in X               | leakage_audit.csv + leakage_audit_full.csv + sample_trace_TB02_24hour.csv | PASS - 0 flagged (180 models audited)                                  |
| A02  | Timestamp & sample counts consistent        | raw_coverage_audit.json + split_statistics.json                           | Reported on observed data (292,447 raw unique ts)                      |
| A03  | Ridge baseline realistic                    | evaluation_metrics.csv                                                    | PASS - Ridge RMSE in line with other models (no R2 ~ 1.0)              |
| A04  | Persistence & Forecast Skill complete       | evaluation_metrics.csv                                                    | PASS - skill_score / skill_vs_ridge populated per row                  |
| A05  | Train/val/test time ranges exact            | split_statistics.json                                                     | Exact start/end per split, no '~'                                      |
| A06  | Model/artifact/API/test counts unified      | inventory_summary.json                                                    | Dynamic counts throughout this report                                  |
| A07  | TB12 missing rate explained                 | data_quality_report.csv + tb12_analysis.json                              | Scoped by column and split                                             |
| A08  | Availability three metrics                  | availability_report.json                                                  | observed / coverage-adjusted / data coverage                           |
| A09  | Farm bias & calibration                     | farm_bias.csv + 25_farm_bias_calibration.png                              | Segmented bias + calibration plot                                      |
| A10  | Ramp/anomaly/failure evidence               | alert_accuracy.csv + anomaly_accuracy.csv                                 | Semantics labeled as heuristic risk scores                             |
| A11  | API security & benchmark                    | tests/test_api.py + logs/api_audit.log                                    | Env-var key, restricted CORS, /health/ alias                           |
| A12  | Report auto-generated, no hardcoded numbers | this PDF + generate_report.py                                             | All tables read from last pipeline run                                 |
| A13  | Reproducible from a clean environment       | README.md + run_all.bat                                                   | run_all.bat: deps -> main.py<br>-> pytest -> generate_report.py -> API |

## Appendix B. Detailed Backtest Tables

Full turbine-level backtest detail referenced from Section 5.9. These wide/tall tables are kept out of the main Results flow to preserve readability.

### B.1 Aggregate metrics across 12 turbines

| Horizon | Model    | MAE          | RMSE         | nRMSE%     | R2 Avg | R2 Min | R2 Max | Bias  | Skill |
|---------|----------|--------------|--------------|------------|--------|--------|--------|-------|-------|
| 10min   | lightgbm | 98.2+/-7.3   | 154.2+/-17.7 | 7.0+/-0.8  | 0.9338 | 0.8645 | 0.9489 | +5.0  | 0.012 |
| 10min   | xgboost  | 99.7+/-7.1   | 155.3+/-17.7 | 7.1+/-0.8  | 0.9330 | 0.8653 | 0.9466 | +6.0  | 0.005 |
| 30min   | lightgbm | 167.3+/-7.7  | 246.3+/-11.4 | 11.2+/-0.5 | 0.8344 | 0.7695 | 0.8594 | +8.1  | 0.055 |
| 30min   | xgboost  | 173.1+/-9.6  | 250.5+/-14.0 | 11.4+/-0.6 | 0.8284 | 0.7496 | 0.8517 | +13.6 | 0.038 |
| 1hour   | lightgbm | 224.1+/-9.0  | 316.1+/-10.7 | 14.4+/-0.5 | 0.7278 | 0.6543 | 0.7635 | +4.5  | 0.082 |
| 1hour   | xgboost  | 230.8+/-10.3 | 320.0+/-12.2 | 14.5+/-0.6 | 0.7208 | 0.6321 | 0.7613 | +12.7 | 0.070 |
| 6hour   | lightgbm | 349.4+/-17.9 | 453.7+/-18.3 | 20.6+/-0.8 | 0.4374 | 0.3101 | 0.5101 | +10.8 | 0.371 |
| 6hour   | xgboost  | 355.6+/-17.9 | 457.8+/-19.4 | 20.8+/-0.9 | 0.4276 | 0.2988 | 0.5049 | +19.1 | 0.366 |
| 24hour  | lightgbm | 405.1+/-17.6 | 511.2+/-15.8 | 23.2+/-0.7 | 0.2883 | 0.1619 | 0.3363 | +15.5 | 0.129 |
| 24hour  | xgboost  | 412.8+/-16.2 | 516.2+/-15.0 | 23.5+/-0.7 | 0.2742 | 0.1483 | 0.3324 | +29.0 | 0.120 |

## B.2 Per-turbine R2 matrix (best model per horizon)

| Turbine     | 10min  | 30min  | 1hour  | 6hour  | 24hour |
|-------------|--------|--------|--------|--------|--------|
| TB01_10min  | 0.9473 | -      | -      | -      | -      |
| TB01_1hour  | -      | -      | 0.7530 | -      | -      |
| TB01_24hour | -      | -      | -      | -      | 0.3423 |
| TB01_30min  | -      | 0.8524 | -      | -      | -      |
| TB01_6hour  | -      | -      | -      | 0.4469 | -      |
| TB02_10min  | 0.9479 | -      | -      | -      | -      |
| TB02_1hour  | -      | -      | 0.7643 | -      | -      |
| TB02_24hour | -      | -      | -      | -      | 0.3849 |
| TB02_30min  | -      | 0.8553 | -      | -      | -      |
| TB02_6hour  | -      | -      | -      | 0.4693 | -      |
| TB03_10min  | 0.9397 | -      | -      | -      | -      |
| TB03_1hour  | -      | -      | 0.7190 | -      | -      |
| TB03_24hour | -      | -      | -      | -      | 0.3458 |
| TB03_30min  | -      | 0.8309 | -      | -      | -      |
| TB03_6hour  | -      | -      | -      | 0.4212 | -      |
| TB04_10min  | 0.9489 | -      | -      | -      | -      |
| TB04_1hour  | -      | -      | 0.7749 | -      | -      |
| TB04_24hour | -      | -      | -      | -      | 0.4009 |
| TB04_30min  | -      | 0.8639 | -      | -      | -      |
| TB04_6hour  | -      | -      | -      | 0.4850 | -      |
| TB05_10min  | 0.9439 | -      | -      | -      | -      |
| TB05_1hour  | -      | -      | 0.7329 | -      | -      |
| TB05_24hour | -      | -      | -      | -      | 0.2649 |
| TB05_30min  | -      | 0.8419 | -      | -      | -      |
| TB05_6hour  | -      | -      | -      | 0.3868 | -      |
| TB06_10min  | 0.9414 | -      | -      | -      | -      |
| TB06_1hour  | -      | -      | 0.7510 | -      | -      |
| TB06_24hour | -      | -      | -      | -      | 0.3805 |
| TB06_30min  | -      | 0.8473 | -      | -      | -      |
| TB06_6hour  | -      | -      | -      | 0.4427 | -      |
| TB07_10min  | 0.9354 | -      | -      | -      | -      |
| TB07_1hour  | -      | -      | 0.7257 | -      | -      |



| Turbine     | 10min  | 30min  | 1hour  | 6hour  | 24hour |
|-------------|--------|--------|--------|--------|--------|
| TB07_24hour | -      | -      | -      | -      | 0.3238 |
| TB07_30min  | -      | 0.8286 | -      | -      | -      |
| TB07_6hour  | -      | -      | -      | 0.4302 | -      |
| TB08_10min  | 0.9438 | -      | -      | -      | -      |
| TB08_1hour  | -      | -      | 0.7616 | -      | -      |
| TB08_24hour | -      | -      | -      | -      | 0.4045 |
| TB08_30min  | -      | 0.8538 | -      | -      | -      |
| TB08_6hour  | -      | -      | -      | 0.4584 | -      |
| TB09_10min  | 0.9438 | -      | -      | -      | -      |
| TB09_1hour  | -      | -      | 0.7509 | -      | -      |
| TB09_24hour | -      | -      | -      | -      | 0.3823 |
| TB09_30min  | -      | 0.8483 | -      | -      | -      |
| TB09_6hour  | -      | -      | -      | 0.5101 | -      |
| TB10_10min  | 0.9399 | -      | -      | -      | -      |
| TB10_1hour  | -      | -      | 0.7473 | -      | -      |
| TB10_24hour | -      | -      | -      | -      | 0.3654 |
| TB10_30min  | -      | 0.8427 | -      | -      | -      |
| TB10_6hour  | -      | -      | -      | 0.4477 | -      |
| TB11_10min  | 0.9394 | -      | -      | -      | -      |
| TB11_1hour  | -      | -      | 0.7449 | -      | -      |
| TB11_24hour | -      | -      | -      | -      | 0.3688 |
| TB11_30min  | -      | 0.8415 | -      | -      | -      |
| TB11_6hour  | -      | -      | -      | 0.4502 | -      |
| TB12_10min  | 0.9310 | -      | -      | -      | -      |
| TB12_1hour  | -      | -      | 0.7219 | -      | -      |
| TB12_24hour | -      | -      | -      | -      | 0.2193 |
| TB12_30min  | -      | 0.8247 | -      | -      | -      |
| TB12_6hour  | -      | -      | -      | 0.3101 | -      |